

Phase 1: Commutative Algebra

The algebraic machinery needed for algebraic number theory and Chebotarev density

Ideals and quotients; Noetherian rings; localization; integral dependence; DVRs; Dedekind domains; fractional ideals and unique factorization of ideals

Purpose. Build enough commutative algebra that prime ideals in rings of integers later feel like concrete arithmetic objects rather than formal notation. Every major theorem is motivated, derived, and followed by exercises. Complete solutions are collected in Part III so the exercises can be attempted honestly first.

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How to use this workbook

Phase 1 is designed for roughly **35–55 focused hours**. The nominal course plan allotted four to five weeks, but you should advance by mastery rather than calendar time.

Recommended cycle for every chapter.

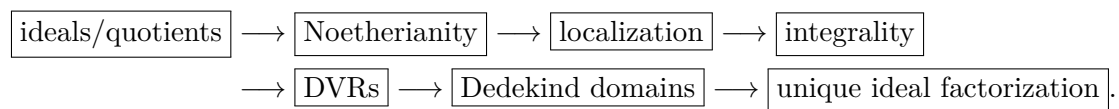
1. Read the conceptual introduction and definitions slowly enough that you can restate them in your own words.
2. Close the text and reconstruct the main proof skeletons on blank paper.
3. Work through the displayed examples yourself before reading the derivation.
4. Attempt the exercises in Part II without consulting Part III.
5. Mark each exercise **C** (confident), **U** (uncertain), or **N** (no viable approach) before starting. This distinguishes mathematical gaps from execution errors.
6. Use the complete solutions diagnostically: identify the first step where your reasoning diverged, then redo the problem from scratch one day later.

Mastery checkpoint

You should leave Phase 1 when you can comfortably explain why localizing at a prime “zooms in” on that prime, why integrality is the correct algebraic analogue of being an algebraic integer, why a Dedekind domain repairs failure of unique factorization of elements, and how unique factorization of ideals is proved rather than merely stated.

Dependency map

The logical spine is



Later, for a number field K , the ring of integers $\mathcal{O}_K$ will be the integral closure of $\mathbb{Z}$ in K . It will be a Dedekind domain. Thus every nonzero ideal admits a unique factorization

$$\mathfrak{a} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r}.$$

Chebotarev ultimately studies how prime ideals $\mathfrak{p}$ behave in finite Galois extensions. Phase 1 builds the language in which that sentence is meaningful.

Part I

Theory and derivations

Chapter 1

Ideals, quotient rings, and prime geometry

1.1 Why ideals are the correct objects

Let A be a commutative ring with $1 \neq 0$. An ideal $I \subseteq A$ is an additive subgroup such that $aI \subseteq I$ for every $a \in A$.

An ideal is not merely a generalized number. It is exactly the kind of subset that can be collapsed to zero while preserving ring operations.

Theorem 1.1 (Kernels are ideals). *If $\varphi : A \rightarrow B$ is a ring homomorphism, then $\ker \varphi$ is an ideal of A .*

Proof. If $x, y \in \ker \varphi$, then

$$\varphi(x - y) = \varphi(x) - \varphi(y) = 0,$$

so $x - y \in \ker \varphi$. If $a \in A$ and $x \in \ker \varphi$, then

$$\varphi(ax) = \varphi(a)\varphi(x) = \varphi(a) \cdot 0 = 0.$$

Hence $ax \in \ker \varphi$. □

Conversely, every ideal occurs as such a kernel: if $I \triangleleft A$, define the quotient ring A/I and the canonical map

$$\pi : A \rightarrow A/I, \quad a \mapsto a + I.$$

Then $\ker \pi = I$.

Intuition

Passing to A/I means declaring every element of I to be zero and imposing all consequences of that declaration. For example, $\mathbb{Z}/(n)$ is arithmetic in which $n = 0$; $k[x, y]/(y - x^2)$ is polynomial arithmetic subject to the relation $y = x^2$.

1.2 The universal property of a quotient

The quotient is characterized not by coset notation but by the maps it permits.

Theorem 1.2 (Universal property). *Let $I \triangleleft A$ and let $f : A \rightarrow B$ be a ring homomorphism with $I \subseteq \ker f$. Then there is a unique ring homomorphism $\bar{f} : A/I \rightarrow B$ satisfying*

$$f = \bar{f} \circ \pi.$$

Explicitly, $\bar{f}(a + I) = f(a)$.

Proof. We must first prove well-definedness. If $a + I = b + I$, then $a - b \in I \subseteq \ker f$, hence $f(a) = f(b)$. Therefore the formula $\bar{f}(a + I) = f(a)$ does not depend on the representative. It is immediately a ring homomorphism. Finally, any map satisfying $f = \bar{f} \circ \pi$ must satisfy $\bar{f}(a + I) = f(a)$, so it is unique. $\square$

1.3 Prime and maximal ideals

Definition 1.3. A proper ideal $\mathfrak{p} \subsetneq A$ is *prime* if

$$ab \in \mathfrak{p} \implies a \in \mathfrak{p} \text{ or } b \in \mathfrak{p}.$$

A proper ideal $\mathfrak{m} \subsetneq A$ is *maximal* if the only ideals containing it are $\mathfrak{m}$ and A .

Theorem 1.4 (Quotient criteria). *For a proper ideal $I \triangleleft A$:*

1. *I is prime if and only if A/I is an integral domain.*
2. *I is maximal if and only if A/I is a field.*

Proof. For (1), $(a + I)(b + I) = 0$ in A/I exactly when $ab \in I$. Thus absence of zero divisors in A/I is exactly the prime-ideal condition.

For (2), suppose I is maximal and $a \notin I$. The ideal $I + (a)$ strictly contains I , hence equals A . So $1 = i + ra$ for some $i \in I$, $r \in A$. Modulo I this gives $(r + I)(a + I) = 1 + I$, so every nonzero coset is invertible. Conversely, if A/I is a field and $I \subseteq J \subseteq A$, then J/I is an ideal of the field A/I , hence either 0 or all of A/I . Therefore $J = I$ or $J = A$. $\square$

Every maximal ideal is prime, because every field is a domain. The converse need not hold: (0) is prime in $\mathbb{Z}$ but not maximal.

1.4 Correspondence of ideals

Theorem 1.5 (Correspondence theorem). *Let $I \triangleleft A$. The map*

$$J \longmapsto J/I$$

is an inclusion-preserving bijection between ideals J of A containing I and ideals of A/I . Under this bijection, prime ideals correspond to prime ideals and maximal ideals correspond to maximal ideals.

Proof. If $J \supseteq I$, then J/I is an ideal of A/I . Conversely, if $K \triangleleft A/I$, then $\pi^{-1}(K)$ is an ideal of A containing I . These constructions are inverse. The prime and maximal statements follow from

$$(A/I)/(J/I) \cong A/J$$

and the quotient criteria above. $\square$

1.5 Sums, products, intersections, and comaximality

For ideals $I, J \triangleleft A$ define

$$I + J = \{i + j : i \in I, j \in J\}, \quad IJ = \left\{ \sum_{r=1}^n i_r j_r : i_r \in I, j_r \in J \right\}.$$

Always $IJ \subseteq I \cap J$. Equality can fail. If $I + J = A$, however, then equality holds.

Proposition 1.6. *If $I + J = A$, then $I \cap J = IJ$.*

Proof. Choose $i \in I$ and $j \in J$ with $i + j = 1$. For $x \in I \cap J$,

$$x = x(i + j) = xi + xj.$$

Here $xi \in IJ = IJ$ because $x \in J$, and $xj \in IJ$ because $x \in I$. Thus $x \in IJ$. $\square$

Theorem 1.7 (Chinese remainder theorem). *If $I_1, \dots, I_n$ are pairwise comaximal ideals, then*

$$A/(I_1 \cdots I_n) \cong A/I_1 \times \cdots \times A/I_n.$$

Equivalently, since pairwise comaximality implies $I_1 \cdots I_n = I_1 \cap \cdots \cap I_n$,

$$A/(I_1 \cap \cdots \cap I_n) \cong \prod_{r=1}^n A/I_r.$$

Proof. It suffices to prove the case $n = 2$ and then induct. Define

$$\Phi : A \rightarrow A/I \times A/J, \quad a \mapsto (a + I, a + J).$$

Its kernel is $I \cap J = IJ$. To prove surjectivity, write $1 = i + j$ with $i \in I, j \in J$. Given $(a + I, b + J)$, set

$$x = aj + bi.$$

Modulo I , $j \equiv 1, i \equiv 0$, so $x \equiv a$. Modulo J , $j \equiv 0, i \equiv 1$, so $x \equiv b$. Thus Φ is surjective. The first isomorphism theorem gives $A/IJ \cong A/I \times A/J$. $\square$

Why this matters for Chebotarev

Later, distinct prime ideals are comaximal in a Dedekind domain. The Chinese remainder theorem lets arithmetic information at finitely many primes be specified independently. This local-to-global flexibility appears constantly in algebraic number theory.

Chapter 2

Noetherian rings: finiteness that survives algebra

2.1 The problem Noetherianity solves

Infinite algebra can become uncontrollable if ideals require endlessly many generators. Noetherianity is a finiteness principle saying that this cannot happen.

Definition 2.1. A ring A is *Noetherian* if every ideal of A is finitely generated.

There are two other equivalent ways to recognize the same property.

Theorem 2.2 (Noetherian equivalences). *For a ring A , the following are equivalent.*

1. *Every ideal of A is finitely generated.*
2. *Every ascending chain of ideals*

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$$

stabilizes.

3. *Every nonempty collection of ideals has a maximal element under inclusion.*

Proof. (1) $\Rightarrow$ (2). Let $I = \bigcup_{n \geq 1} I_n$. Because the chain is ascending, I is an ideal. By (1), $I = (a_1, \dots, a_r)$. Every a_j belongs to some I_{N_j} . Taking $N = \max N_j$, all generators lie in I_N , so $I \subseteq I_N$. The reverse inclusion is automatic, hence $I = I_N$, and therefore $I_n = I_N$ for all $n \geq N$.

(2) $\Rightarrow$ (3). Suppose a nonempty family $\mathcal{F}$ of ideals had no maximal member. Choose $I_1 \in \mathcal{F}$. Since it is not maximal, choose $I_2 \in \mathcal{F}$ with $I_1 \subsetneq I_2$, and continue. This gives a strictly ascending chain, contradicting (2).

(3) $\Rightarrow$ (1). Let $I \triangleleft A$. Consider the family of finitely generated ideals contained in I . It is nonempty because (0) belongs to it. Choose a maximal member $J = (a_1, \dots, a_r)$. If $J \neq I$, choose $x \in I \setminus J$. Then $J + (x)$ is a finitely generated ideal contained in I and strictly larger than J , contradiction. Hence $J = I$. $\square$

Intuition

A Noetherian ring cannot keep revealing genuinely new ideal-theoretic information forever. Any process that repeatedly enlarges an ideal must eventually stop. This is why maximal-counterexample arguments work so well in Noetherian algebra.

2.2 Modules and the finite-module principle

A brief amount of module language makes later integrality proofs dramatically cleaner.

Definition 2.3. An A -module M is *finitely generated* if there exist $m_1, \dots, m_r \in M$ such that every $m \in M$ can be written

$$m = a_1 m_1 + \dots + a_r m_r$$

with $a_i \in A$.

Theorem 2.4. *If A is Noetherian and M is a finitely generated A -module, then every submodule of M is finitely generated.*

Proof. It is enough to prove the claim for $M = A^n$ by induction on n , because a finite generating set gives a surjection $A^n \twoheadrightarrow M$ and submodules pull back to submodules.

For $n = 1$ the statement is exactly that ideals of A are finitely generated. Assume it for A^{n-1} and let $N \subseteq A^n$. Project onto the last coordinate:

$$\pi : N \rightarrow A.$$

The image $I = \pi(N)$ is an ideal, hence generated by $a_1, \dots, a_r$. Choose $v_i \in N$ with last coordinate a_i . The kernel $K = N \cap A^{n-1}$ is finitely generated by the induction hypothesis, say by $w_1, \dots, w_s$. If $v \in N$, its last coordinate is $\sum c_i a_i$, so

$$v - \sum c_i v_i \in K.$$

Thus the v_i and w_j generate N . □

Corollary 2.5. *If A is Noetherian and B is a ring that is finitely generated as an A -module, then B is Noetherian.*

Proof. Any ideal $I \triangleleft B$ is in particular an A -submodule of the finite A -module B , hence is generated over A by finitely many elements $x_1, \dots, x_r$. The same elements generate I as a B -ideal because $A \subseteq B$. □

2.3 Quotients and polynomial rings

Proposition 2.6. *If A is Noetherian and $I \triangleleft A$, then A/I is Noetherian.*

Proof. Ideals of A/I correspond to ideals of A containing I . If J/I is such an ideal and $J = (a_1, \dots, a_r)$, then J/I is generated by the cosets $a_i + I$. □

Theorem 2.7 (Hilbert basis theorem). *If A is Noetherian, then $A[x]$ is Noetherian.*

Proof. Let $I \triangleleft A[x]$. Let $J \subseteq A$ be the ideal generated by the leading coefficients of all nonzero polynomials in I . Since A is Noetherian, choose generators

$$J = (a_1, \dots, a_r),$$

and for each a_i choose $f_i \in I$ having leading coefficient a_i . Let

$$N = \max_i \deg f_i.$$

Now let $M \subseteq I$ be the set of polynomials of degree $< N$ (including 0). Via the coefficient map

$$c_0 + c_1x + \dots + c_{N-1}x^{N-1} \mapsto (c_0, \dots, c_{N-1}),$$

M is an A -submodule of A^N . Because A is Noetherian, every submodule of the finite module A^N is finitely generated. Choose $g_1, \dots, g_s$ generating M as an A -module.

We claim that

$$I = (f_1, \dots, f_r, g_1, \dots, g_s).$$

Let $f \in I$ and induct on $d = \deg f$. If $d < N$, then $f \in M$ and is an A -linear combination of the g_j . If $d \geq N$, let a be the leading coefficient of f . Since $a \in J$, write

$$a = c_1a_1 + \dots + c_ra_r.$$

Because $d \geq N \geq \deg f_i$, the polynomial

$$h = f - \sum_{i=1}^r c_i x^{d-\deg f_i} f_i$$

lies in I and has degree strictly smaller than d : the displayed sum has exactly the same leading term ax^d as f . By induction, h lies in the ideal generated by the f_i and g_j , and therefore so does f . Hence I is finitely generated. $\square$

Remark 2.8. The bookkeeping in the preceding proof is less important than the mechanism: Noetherianity of coefficients prevents new leading-coefficient ideals from appearing indefinitely, and then induction on degree finishes the job.

Example 2.9. By repeated application, $k[x_1, \dots, x_n]$ is Noetherian for every field k and finite n . In contrast, $k[x_1, x_2, \dots]$ is not Noetherian because

$$(x_1) \subsetneq (x_1, x_2) \subsetneq (x_1, x_2, x_3) \subsetneq \dots$$

does not stabilize.

Why this matters for Chebotarev

The ring of integers $\mathcal{O}_K$ of a number field is a finitely generated $\mathbb{Z}$ -module. Since $\mathbb{Z}$ is Noetherian, the finite-module principle will imply that $\mathcal{O}_K$ is Noetherian. This supplies one of the three defining properties of a Dedekind domain.

Chapter 3

Localization: algebraic zoom at a prime

3.1 The construction

Let A be a ring and $S \subseteq A$ a multiplicatively closed subset: $1 \in S$ and $s, t \in S$ imply $st \in S$. We want a ring in which every element of S becomes invertible.

Consider pairs $(a, s) \in A \times S$. Define

$$(a, s) \sim (b, t)$$

if there exists $u \in S$ such that

$$u(at - bs) = 0.$$

The factor u is necessary when A has zero divisors. The equivalence class of (a, s) is written a/s .

Define

$$\frac{a}{s} + \frac{b}{t} = \frac{at + bs}{st}, \quad \frac{a}{s} \frac{b}{t} = \frac{ab}{st}.$$

The resulting ring is the *localization* $S^{-1}A$.

If A is a domain and $0 \notin S$, the relation simplifies to the familiar cross-multiplication condition $at = bs$.

3.2 Universal property

Theorem 3.1 (Universal property of localization). *Let $f : A \rightarrow B$ be a ring homomorphism such that $f(s)$ is a unit of B for every $s \in S$. Then there is a unique ring homomorphism*

$$\tilde{f} : S^{-1}A \rightarrow B$$

satisfying $\tilde{f}(a/1) = f(a)$. It is given by

$$\tilde{f}(a/s) = f(a)f(s)^{-1}.$$

Proof. The formula is forced because $a/s = (a/1)(s/1)^{-1}$. To prove it is well defined, suppose $a/s = b/t$. Then for some $u \in S$, $u(at - bs) = 0$. Applying f gives

$$f(u)(f(a)f(t) - f(b)f(s)) = 0.$$

Since $f(u)$ is invertible,

$$f(a)f(t) = f(b)f(s).$$

Multiplying by $f(s)^{-1}f(t)^{-1}$ gives the required equality. Ring-homomorphism properties are immediate. $\square$

Intuition

Localization is the smallest possible modification of A that makes a chosen collection S invertible. It does not add unrelated relations. This universal characterization is more important than the fraction notation.

3.3 Important localizations

3.3.1 Inverting one element

For $f \in A$, let $S = \{1, f, f^2, \dots\}$. Write

$$A_f = S^{-1}A.$$

Example: $\mathbb{Z}_6$ in this notation would be ambiguous, so write $\mathbb{Z}[1/6]$ for the localization obtained by inverting powers of 6. Its elements are rational numbers whose denominators divide a power of 6.

3.3.2 Localization at a prime

Let $\mathfrak{p}$ be prime and take

$$S = A \setminus \mathfrak{p}.$$

This is multiplicatively closed precisely because $\mathfrak{p}$ is prime. Write

$$A_{\mathfrak{p}} = (A \setminus \mathfrak{p})^{-1}A.$$

Every element outside $\mathfrak{p}$ becomes a unit.

Proposition 3.2. $A_{\mathfrak{p}}$ is a local ring: it has a unique maximal ideal, namely

$$\mathfrak{p}A_{\mathfrak{p}} = \left\{ \frac{a}{s} : a \in \mathfrak{p}, s \notin \mathfrak{p} \right\}.$$

Proof. If $a/s \notin \mathfrak{p}A_{\mathfrak{p}}$, then $a \notin \mathfrak{p}$. Hence $a \in A \setminus \mathfrak{p}$ and

$$\frac{a}{s} \cdot \frac{s}{a} = 1,$$

so a/s is a unit. Thus every element outside $\mathfrak{p}A_{\mathfrak{p}}$ is a unit. Every proper ideal contains no unit, so every proper ideal is contained in $\mathfrak{p}A_{\mathfrak{p}}$. Hence $\mathfrak{p}A_{\mathfrak{p}}$ is the unique maximal ideal. $\square$

Intuition

In $A_{\mathfrak{p}}$, anything not divisible by $\mathfrak{p}$ has been declared irrelevant and made invertible. What remains noninvertible is exactly the arithmetic controlled by $\mathfrak{p}$. This is the algebraic version of zooming in on one prime.

3.4 Prime ideals under localization

Theorem 3.3 (Prime correspondence for localization). *Prime ideals of $S^{-1}A$ correspond bijectively to prime ideals $\mathfrak{p}$ of A satisfying $\mathfrak{p} \cap S = \emptyset$. The correspondence is*

$$\mathfrak{p} \longmapsto S^{-1}\mathfrak{p},$$

with inverse contraction along $A \rightarrow S^{-1}A$.

Proof. If $\mathfrak{p} \cap S = \emptyset$, then $S^{-1}\mathfrak{p}$ is proper. Indeed, if $1 = a/s$ with $a \in \mathfrak{p}$, then for some $u \in S$ we would have $u(s - a) = 0$, hence $us = ua \in \mathfrak{p}$. But $u, s \in S$, so $us \in S$, contradicting $\mathfrak{p} \cap S = \emptyset$. To see primality, if $(a/s)(b/t) \in S^{-1}\mathfrak{p}$, then after clearing denominators there is $u \in S$ with $uab \in \mathfrak{p}$. Since $u \notin \mathfrak{p}$ and $\mathfrak{p}$ is prime, $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$, so one factor belongs to $S^{-1}\mathfrak{p}$.

Conversely, if $Q \subseteq S^{-1}A$ is prime, its contraction $\mathfrak{p}$ is prime. No $s \in S$ can lie in $\mathfrak{p}$, because $s/1$ is a unit and a proper prime ideal cannot contain a unit. Finally one verifies extension and contraction are inverse: every fraction in Q has numerator in the contraction because its denominator is a unit. $\square$

For $A_{\mathfrak{p}}$, the surviving primes are exactly the primes $\mathfrak{q} \subseteq \mathfrak{p}$.

3.5 Localization of ideals and modules

If $I \triangleleft A$, its extension is

$$S^{-1}I = \left\{ \frac{a}{s} : a \in I, s \in S \right\}.$$

More generally, for an A -module M one forms $S^{-1}M$ from symbols m/s with the analogous equivalence relation.

A useful practical fact is that localization commutes with finite sums, products, and quotients in the expected way. In particular,

$$S^{-1}(A/I) \cong S^{-1}A/S^{-1}I.$$

3.6 Noetherianity localizes

Proposition 3.4. *If A is Noetherian, then $S^{-1}A$ is Noetherian.*

Proof. Let $J \triangleleft S^{-1}A$ and contract it to $I \triangleleft A$. Every element $a/s \in J$ satisfies $a/1 = (s/1)(a/s) \in J$, so $a \in I$ and hence $a/s \in S^{-1}I$. Thus $J = S^{-1}I$. Since $I = (a_1, \dots, a_r)$, the ideal J is generated by $a_1/1, \dots, a_r/1$. $\square$

Example 3.5 (The integers localized at (p)). For a rational prime p ,

$$\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} \in \mathbb{Q} : p \nmid b \right\}.$$

Its unique maximal ideal is

$$p\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} : p \mid a, p \nmid b \right\}.$$

Every nonzero $x \in \mathbb{Q}$ can be written uniquely as

$$x = p^n u$$

where $n \in \mathbb{Z}$ and u has numerator and denominator both prime to p . Inside $\mathbb{Z}_{(p)}$, u is a unit and $n \geq 0$. This is the prototype of a discrete valuation ring.

Why this matters for Chebotarev

For a number field K and a prime ideal $\mathfrak{p} \subset \mathcal{O}_K$, the local ring $(\mathcal{O}_K)_{\mathfrak{p}}$ isolates the arithmetic of that one prime. In a Dedekind domain it will be a DVR, meaning that every nonzero ideal locally is simply a power of $\mathfrak{p}$.

Chapter 4

Integral dependence and integral closure

4.1 Why monic polynomials matter

Let $A \subseteq B$ be rings.

Definition 4.1. An element $b \in B$ is *integral over* A if it satisfies a monic polynomial

$$b^n + a_{n-1}b^{n-1} + \cdots + a_1b + a_0 = 0, \quad a_i \in A.$$

The extension B/A is *integral* if every element of B is integral over A .

The adjective “monic” is the source of finiteness. Rearranging gives

$$b^n = -a_{n-1}b^{n-1} - \cdots - a_0,$$

so every higher power of b reduces to an A -linear combination of

$$1, b, \dots, b^{n-1}.$$

Therefore $A[b]$ is a finite A -module.

4.2 The finite-module criterion

Theorem 4.2 (Finite-module criterion and determinant trick). *For $b \in B$, the following are equivalent.*

1. b is integral over A .
2. The ring $A[b]$ is finitely generated as an A -module.
3. There exists a finitely generated A -submodule $M \subseteq B$ such that $1 \in M$ and $bM \subseteq M$.

Proof. (1) $\Rightarrow$ (2) was just explained: if b satisfies a monic equation of degree n , then $1, b, \dots, b^{n-1}$ generate $A[b]$.

(2) $\Rightarrow$ (3): take $M = A[b]$. It contains 1, is finitely generated by assumption, and multiplication by b preserves it.

(3) $\Rightarrow$ (1): let $m_1, \dots, m_r$ generate M . Since $bM \subseteq M$, for each i there exist $a_{ij} \in A$ such that

$$bm_i = \sum_{j=1}^r a_{ij}m_j.$$

In matrix form,

$$(bI_r - C)\mathbf{m} = 0,$$

where $C = (a_{ij})$ and $\mathbf{m} = (m_1, \dots, m_r)^T$. Multiply by the adjugate matrix:

$$\det(bI_r - C)\mathbf{m} = 0.$$

Thus the monic polynomial $P(T) = \det(TI_r - C) \in A[T]$ satisfies $P(b)m = 0$ for every $m \in M$. Because $1 \in M$, we may take $m = 1$ and obtain

$$P(b) = P(b) \cdot 1 = 0.$$

Hence b is integral over A . □

Intuition

An integral element is algebraic in a way that does not create an infinite new direction over the base ring. Adjoining it produces a module of finite rank. This is why algebraic integers behave like finite-dimensional arithmetic objects.

4.3 Integral elements form a ring

Theorem 4.3. *If $x, y \in B$ are integral over A , then $x + y$ and xy are integral over A . Thus the elements of B integral over A form a subring.*

Proof. Because x is integral, $A[x]$ is finite over A . Because y is integral over A , it is also integral over the larger ring $A[x]$ (the same monic equation still has coefficients there). Hence $A[x, y] = A[x][y]$ is finite as an $A[x]$ -module. A module finite over a finite A -module is finite over A : products of generating sets generate. Therefore $A[x, y]$ is finite over A .

Multiplication by either $x + y$ or xy preserves the finite A -module $A[x, y]$, which contains 1. The determinant trick therefore gives a monic equation for each element, so both are integral. □

4.4 Transitivity of integrality

Theorem 4.4. *If $A \subseteq B \subseteq C$, B is integral over A , and $c \in C$ is integral over B , then c is integral over A . Consequently an integral extension of an integral extension is integral.*

Proof. Suppose

$$c^n + b_{n-1}c^{n-1} + \dots + b_0 = 0$$

with $b_i \in B$. Only finitely many coefficients occur. Let

$$B_0 = A[b_0, \dots, b_{n-1}].$$

Each b_i is integral over A , so repeated use of the finite-module criterion shows B_0 is finite over A . Since c is integral over B_0 , $B_0[c]$ is finite over B_0 , hence finite over A . The determinant trick implies c is integral over A . $\square$

4.5 Integral closure

Definition 4.5. If A is a domain with fraction field K , the *integral closure* of A in an extension field L/K is the set of elements of L integral over A . The domain A is *integrally closed* if every element of K integral over A already lies in A .

Theorem 4.6. *Every UFD is integrally closed.*

Proof. Let A be a UFD with fraction field K , and suppose $x = a/b \in K$ is integral, with $a, b \in A$ chosen coprime. A monic equation gives

$$\left(\frac{a}{b}\right)^n + c_{n-1}\left(\frac{a}{b}\right)^{n-1} + \dots + c_0 = 0.$$

Multiplying by b^n ,

$$a^n = -b(c_{n-1}a^{n-1} + c_{n-2}a^{n-2}b + \dots + c_0b^{n-1}).$$

Thus $b \mid a^n$. If an irreducible π divided b , then because irreducibles are prime in a UFD, $\pi \mid a^n$ would imply $\pi \mid a$, contradicting coprimality. Therefore b is a unit and $x \in A$. $\square$

In particular, $\mathbb{Z}$ and $k[t]$ are integrally closed.

Example 4.7 (A first algebraic integer). Let d be squarefree. The element $\sqrt{d}$ satisfies $T^2 - d = 0$, so it is integral over $\mathbb{Z}$. If $d \equiv 1 \pmod{4}$, then

$$\alpha = \frac{1 + \sqrt{d}}{2}$$

satisfies

$$\alpha^2 - \alpha + \frac{1-d}{4} = 0,$$

and $(1-d)/4 \in \mathbb{Z}$. Hence α is also integral. This foreshadows the formula for rings of integers of quadratic fields.

4.6 Integrality and localization

Proposition 4.8. *Let $A \subseteq B$ and let $S \subseteq A$ be multiplicatively closed. If $b \in B$ is integral over A , then $b/1$ is integral over $S^{-1}A$. More generally, $S^{-1}B$ is integral over $S^{-1}A$ whenever B is integral over A .*

Proof. A monic equation for b over A remains a monic equation after applying the localization map to its coefficients. For an arbitrary $b/s \in S^{-1}B$, write it as $(b/1)(1/s)$. Since $1/s \in S^{-1}A$, multiplication by it preserves integrality: if b satisfies a monic degree- n equation, then b/s satisfies the equation obtained by dividing the coefficient of T^i by the appropriate power of s . $\square$

Why this matters for Chebotarev

For a number field K , the ring of integers $\mathcal{O}_K$ is the integral closure of $\mathbb{Z}$ in K . The word “integer” is therefore not metaphorical: it means precisely “integral over $\mathbb{Z}$.” Integral closure is the second defining property of a Dedekind domain.

Chapter 5

Factorization domains and discrete valuation rings

5.1 PID and UFD: two forms of factorization

A principal ideal domain (PID) is an integral domain in which every ideal is principal. A unique factorization domain (UFD) is a domain in which every nonzero nonunit factors into irreducibles uniquely up to order and associates.

Theorem 5.1. *Every PID is a UFD.*

Proof. First prove existence. If a nonzero nonunit a cannot be factored into irreducibles, then $a = a_1 b_1$ with both factors nonunits and $(a) \subsetneq (a_1)$. Repeating would create a strictly ascending chain of principal ideals

$$(a) \subsetneq (a_1) \subsetneq (a_2) \subsetneq \cdots,$$

which cannot continue because a PID is Noetherian. Thus factorization into irreducibles exists.

For uniqueness it suffices to show irreducibles are prime. Let π be irreducible and suppose $\pi \mid ab$. The ideal (π, a) is principal, say (d) . Since $d \mid \pi$, irreducibility implies d is a unit or associate to π . If d is associate to π , then $\pi \mid a$. If d is a unit, Bezout gives $r\pi + sa = 1$; multiplying by b gives $b = r\pi b + sab$, both terms on the right divisible by π , hence $\pi \mid b$. Thus π is prime, and the standard cancellation argument gives uniqueness. $\square$

The converses fail. For example, $k[x, y]$ is a UFD but not a PID.

5.2 Valuations

Definition 5.2. A *discrete valuation* on a field K is a surjective map

$$v : K^\times \rightarrow \mathbb{Z}$$

such that

$$v(xy) = v(x) + v(y), \quad v(x + y) \geq \min\{v(x), v(y)\}$$

whenever $x + y \neq 0$. Set $v(0) = +\infty$.

Associated to v is the valuation ring

$$A_v = \{x \in K : v(x) \geq 0\},$$

with maximal ideal

$$\mathfrak{m}_v = \{x : v(x) > 0\}.$$

Choose a *uniformizer* π with $v(\pi) = 1$.

Proposition 5.3. *Every nonzero $x \in A_v$ has a unique expression*

$$x = u\pi^n,$$

where $u \in A_v^\times$ and $n = v(x) \geq 0$.

Proof. Set $n = v(x)$. Then $u = x\pi^{-n}$ has valuation 0. Both u and u^{-1} have nonnegative valuation, so u is a unit. Uniqueness follows by applying v . $\square$

Theorem 5.4 (Ideals in a DVR). *Every nonzero ideal I of a discrete valuation ring is of the form*

$$I = (\pi^n) = \mathfrak{m}^n$$

for a unique $n \geq 0$.

Proof. The set $\{v(x) : 0 \neq x \in I\} \subseteq \mathbb{Z}_{\geq 0}$ has a least element n . Choose $x \in I$ with $v(x) = n$. Then $x = u\pi^n$, so $(x) = (\pi^n) \subseteq I$. Conversely, every $y \in I$ has $v(y) \geq n$, hence y/x has nonnegative valuation and lies in the ring. Thus $y \in (x)$. Therefore $I = (\pi^n)$. $\square$

5.3 Intrinsic characterization of a DVR

Theorem 5.5. *Let $(A, \mathfrak{m})$ be a Noetherian local domain that is not a field. If $\mathfrak{m}$ is principal, then A is a DVR.*

Proof. Write $\mathfrak{m} = (\pi)$. Let $0 \neq x \in A$. If x is not a unit, then $x \in \mathfrak{m}$, so $x = \pi x_1$. If x_1 is not a unit, continue. This process must terminate: otherwise

$$(x) \subsetneq (x_1) \subsetneq (x_2) \subsetneq \cdots$$

would be an infinite strictly ascending chain of ideals. Hence

$$x = u\pi^n$$

for some unit u and $n \geq 0$. Extend to the fraction field by $v(a/b) = v(a) - v(b)$. The factorization shows this is well defined and gives a discrete valuation whose valuation ring is A . $\square$

Example 5.6. $\mathbb{Z}_{(p)}$ is a DVR with uniformizer p and valuation v_p . Similarly, $k[t]_{(t)}$ is a DVR with uniformizer t .

Intuition

A DVR is one-dimensional arithmetic with only one prime direction. Every nonzero element is “a unit times a power of the prime.” It is the local analogue of the elementary statement that an integer has a well-defined exponent of p in its factorization.

Chapter 6

Dedekind domains

6.1 Definition and first examples

Definition 6.1. A *Dedekind domain* is an integral domain A satisfying:

1. A is Noetherian;
2. A is integrally closed in its fraction field;
3. every nonzero prime ideal is maximal.

The last property is often called Krull dimension one.

Every PID is a Dedekind domain: it is Noetherian; it is a UFD and therefore integrally closed; and every nonzero prime ideal (p) is maximal because $A/(p)$ is a domain and, in a PID, irreducibility of p implies the quotient is a field.

Thus $\mathbb{Z}$ and $k[t]$ are Dedekind domains.

6.2 The local characterization

The reason Dedekind domains are tractable is that locally they are DVRs.

Theorem 6.2 (One-dimensional normal local rings). *Let $(A, \mathfrak{m})$ be a Noetherian local domain with fraction field K . Assume A is integrally closed and its only prime ideals are (0) and $\mathfrak{m}$. If A is not a field, then A is a DVR.*

Proof. Choose $0 \neq x \in \mathfrak{m}$. Since every prime ideal containing (x) is nonzero and therefore equals $\mathfrak{m}$, we have

$$\sqrt{(x)} = \mathfrak{m}.$$

Because $\mathfrak{m}$ is finitely generated, say $\mathfrak{m} = (a_1, \dots, a_r)$, and each $a_i \in \sqrt{(x)}$, there are exponents n_i with $a_i^{n_i} \in (x)$. Taking $N = n_1 + \dots + n_r$, every monomial of total degree N in the a_i contains some $a_i^{n_i}$, so

$$\mathfrak{m}^N \subseteq (x).$$

Choose N minimal with this property. Then $\mathfrak{m}^{N-1} \not\subseteq (x)$, so choose

$$y \in \mathfrak{m}^{N-1} \setminus (x).$$

For every $z \in \mathfrak{m}$, $yz \in \mathfrak{m}^N \subseteq (x)$, hence

$$t := y/x \in K$$

satisfies $t\mathfrak{m} \subseteq A$. Since $y \notin (x)$, $t \notin A$.

We claim $t\mathfrak{m} \not\subseteq \mathfrak{m}$. If instead $t\mathfrak{m} \subseteq \mathfrak{m}$, choose generators $m_1, \dots, m_r$ of the nonzero ideal $\mathfrak{m}$. Writing $tm_i = \sum_j a_{ij}m_j$ and applying the same adjugate-matrix argument as above produces a monic polynomial $P(T) \in A[T]$ such that $P(t)m_i = 0$ for every i . Choose a nonzero generator m_i . Since A is a domain and t lies in its fraction field, $P(t)m_i = 0$ implies $P(t) = 0$. Thus t is integral over A . Integral closedness would force $t \in A$, contradiction.

Therefore there exists $z \in \mathfrak{m}$ such that $tz \notin \mathfrak{m}$. In a local ring, elements outside the maximal ideal are units, so $u := tz$ is a unit. Thus

$$t^{-1} = zu^{-1} \in \mathfrak{m}.$$

For any $w \in \mathfrak{m}$, $tw \in A$, so

$$w = t^{-1}(tw) \in t^{-1}A.$$

Hence $\mathfrak{m} \subseteq t^{-1}A$. The reverse inclusion holds because $t^{-1} \in \mathfrak{m}$. Therefore

$$\mathfrak{m} = t^{-1}A$$

is principal. By the preceding characterization, A is a DVR. □

Corollary 6.3 (Local characterization of Dedekind domains). *If A is a Dedekind domain and $0 \neq \mathfrak{p} \subset A$ is prime, then $A_{\mathfrak{p}}$ is a DVR.*

Proof. Localization preserves Noetherianity. It also preserves integral closedness. To see this explicitly, suppose $x \in \text{Frac}(A)$ is integral over $A_{\mathfrak{p}}$ and satisfies

$$x^n + c_{n-1}x^{n-1} + \dots + c_0 = 0, \quad c_i \in A_{\mathfrak{p}}.$$

Choose one element $s \notin \mathfrak{p}$ that is a common multiple of all denominators appearing among the finitely many c_i . Set $y = sx$. Multiplying the equation by s^n gives

$$y^n + (sc_{n-1})y^{n-1} + (s^2c_{n-2})y^{n-2} + \dots + s^nc_0 = 0,$$

and each displayed coefficient lies in A after choosing s large enough (replace s by a product of the denominators if necessary). Hence y is integral over A . Since A is integrally closed, $y \in A$, and therefore $x = y/s \in A_{\mathfrak{p}}$.

The prime ideals of $A_{\mathfrak{p}}$ correspond to prime ideals of A contained in $\mathfrak{p}$. Since every nonzero prime of A is maximal, the only such primes are (0) and $\mathfrak{p}$. Apply the theorem. □

6.3 Fractional ideals

Let A be a domain with fraction field K .

Definition 6.4. A *fractional ideal* is a nonzero A -submodule $I \subseteq K$ such that there exists $0 \neq d \in A$ with $dI \subseteq A$.

Ordinary nonzero ideals are fractional ideals. So are objects such as $(1/6)(2, 3)$ inside $\mathbb{Q}$.

For fractional ideals I, J , define their product as the A -submodule generated by products xy with $x \in I, y \in J$.

Define

$$I^{-1} = \{x \in K : xI \subseteq A\}.$$

If $II^{-1} = A$, then I is *invertible*.

6.4 Every nonzero ideal in a Dedekind domain is invertible

We need two elementary localization facts.

Lemma 6.5. *Let I be a finitely generated fractional ideal and $\mathfrak{p}$ a prime. Then*

$$(I^{-1})_{\mathfrak{p}} = (I_{\mathfrak{p}})^{-1}.$$

Proof. The inclusion $(I^{-1})_{\mathfrak{p}} \subseteq (I_{\mathfrak{p}})^{-1}$ is immediate. For the reverse inclusion, write $I = (a_1, \dots, a_r)$ after clearing a common denominator. Suppose $x \in (I_{\mathfrak{p}})^{-1}$. Then for each i , $xa_i \in A_{\mathfrak{p}}$, so there exists $s_i \notin \mathfrak{p}$ with $s_i xa_i \in A$. Let $s = s_1 \cdots s_r \notin \mathfrak{p}$. Then $sxI \subseteq A$, so $sx \in I^{-1}$ and

$$x = (sx)/s \in (I^{-1})_{\mathfrak{p}}.$$

□

Lemma 6.6 (Local equality detects global equality). *If $J \subseteq A$ is an ideal and $J_{\mathfrak{m}} = A_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$, then $J = A$.*

Proof. If $J \neq A$, it is contained in some maximal ideal $\mathfrak{m}$. Then $J_{\mathfrak{m}} \subseteq \mathfrak{m}A_{\mathfrak{m}}$, so $J_{\mathfrak{m}} \neq A_{\mathfrak{m}}$, contradiction. □

Theorem 6.7. *Every nonzero fractional ideal of a Dedekind domain is invertible.*

Proof. It suffices after scaling to treat an ordinary nonzero ideal I . Since A is Noetherian, I is finitely generated. For every maximal ideal $\mathfrak{p}$, the localization $A_{\mathfrak{p}}$ is a DVR, so

$$I_{\mathfrak{p}} = (\pi^{n_{\mathfrak{p}}})$$

for some $n_{\mathfrak{p}} \geq 0$. Therefore $I_{\mathfrak{p}}$ is invertible and

$$I_{\mathfrak{p}}(I_{\mathfrak{p}})^{-1} = A_{\mathfrak{p}}.$$

Using the previous lemma,

$$(II^{-1})_{\mathfrak{p}} = I_{\mathfrak{p}}(I^{-1})_{\mathfrak{p}} = I_{\mathfrak{p}}(I_{\mathfrak{p}})^{-1} = A_{\mathfrak{p}}$$

for every maximal $\mathfrak{p}$. Since $II^{-1} \subseteq A$, the local-equality lemma gives $II^{-1} = A$. □

6.5 Unique factorization of ideals

We can now prove the central theorem of Phase 1.

Theorem 6.8 (Unique factorization of ideals). *Let A be a Dedekind domain. Every nonzero proper ideal I can be written*

$$I = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r}$$

with distinct nonzero prime ideals $\mathfrak{p}_i$ and positive integers e_i . This factorization is unique up to ordering.

Proof. Existence. Suppose, for contradiction, that there are nonzero proper ideals that are not products of prime ideals. Since A is Noetherian, the collection of such counterexamples has a maximal element I under inclusion. Choose a maximal ideal $\mathfrak{p}$ containing I . Since I and $\mathfrak{p}$ are invertible, set

$$J = \mathfrak{p}^{-1}I.$$

Because $I \subseteq \mathfrak{p}$, multiplying by $\mathfrak{p}^{-1}$ gives $J \subseteq A$, so J is an ordinary ideal. Also

$$I = \mathfrak{p}J.$$

We claim $I \subsetneq J$. Certainly $I \subseteq J$ because $A \subseteq \mathfrak{p}^{-1}$. If equality held, $I = \mathfrak{p}I$; multiplying by I^{-1} would give $A = \mathfrak{p}$, impossible. Hence J strictly contains I . By maximality of the counterexample I , the ideal J is a product of prime ideals. Then $I = \mathfrak{p}J$ is also a product of prime ideals, contradiction.

Uniqueness. Suppose

$$\mathfrak{p}_1 \cdots \mathfrak{p}_r = \mathfrak{q}_1 \cdots \mathfrak{q}_s$$

where all factors are nonzero prime ideals, repetitions allowed. The right-hand product is contained in $\mathfrak{p}_1$. If none of the $\mathfrak{q}_j$ were contained in $\mathfrak{p}_1$, choose $x_j \in \mathfrak{q}_j \setminus \mathfrak{p}_1$; then $x_1 \cdots x_s \in \mathfrak{q}_1 \cdots \mathfrak{q}_s \subseteq \mathfrak{p}_1$, contradicting primality of $\mathfrak{p}_1$. Thus some $\mathfrak{q}_j \subseteq \mathfrak{p}_1$. Both are nonzero prime ideals and hence maximal, so $\mathfrak{q}_j = \mathfrak{p}_1$. Cancel this common invertible factor and proceed inductively. The multisets of prime factors coincide. $\square$

Intuition

The proof separates into two ideas. Noetherianity supplies a maximal counterexample, while invertibility lets us divide an ideal by a prime ideal and cancel factors. This is the ideal-theoretic replacement for ordinary integer factorization.

6.6 Valuations attached to prime ideals

For each nonzero prime $\mathfrak{p}$, the local ring $A_{\mathfrak{p}}$ is a DVR. Let $v_{\mathfrak{p}}$ denote its normalized valuation. If

$$I = \prod_{\mathfrak{p}} \mathfrak{p}^{e_{\mathfrak{p}}},$$

where only finitely many $e_{\mathfrak{p}}$ are nonzero, define

$$v_{\mathfrak{p}}(I) = e_{\mathfrak{p}}.$$

Then

$$v_{\mathfrak{p}}(IJ) = v_{\mathfrak{p}}(I) + v_{\mathfrak{p}}(J).$$

For a nonzero element $x \in K^\times$, apply this to the principal fractional ideal (x) .

This is a major conceptual simplification: ideal multiplication becomes addition of integer exponents, one prime at a time.

6.7 PID versus Dedekind: what has been repaired?

A Dedekind domain need not be a UFD. The difference is that element factorization may fail while ideal factorization remains perfect.

Example 6.9. The ring $\mathbb{Z}[\sqrt{-5}]$. Let

$$A = \mathbb{Z}[\sqrt{-5}].$$

Later you will prove this is the full ring of integers of $\mathbb{Q}(\sqrt{-5})$, hence a Dedekind domain. At the element level,

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5}),$$

and these factorizations cannot be reconciled by units and associates.

At the ideal level define

$$\mathfrak{p}_2 = (2, 1 + \sqrt{-5}), \quad \mathfrak{p}_3 = (3, 1 + \sqrt{-5}), \quad \bar{\mathfrak{p}}_3 = (3, 1 - \sqrt{-5}).$$

The quotients $A/\mathfrak{p}_2 \cong \mathbb{F}_2$ and $A/\mathfrak{p}_3 \cong A/\bar{\mathfrak{p}}_3 \cong \mathbb{F}_3$ show these are maximal ideals. One verifies

$$(2) = \mathfrak{p}_2^2, \quad (3) = \mathfrak{p}_3 \bar{\mathfrak{p}}_3,$$

and

$$(1 + \sqrt{-5}) = \mathfrak{p}_2 \mathfrak{p}_3, \quad (1 - \sqrt{-5}) = \mathfrak{p}_2 \bar{\mathfrak{p}}_3.$$

Thus both element factorizations of 6 induce the same unique ideal factorization

$$(6) = \mathfrak{p}_2^2 \mathfrak{p}_3 \bar{\mathfrak{p}}_3.$$

The apparent contradiction disappears after passing from elements to ideals.

Why this matters for Chebotarev

In algebraic number theory, a rational prime p gives the principal ideal (p) in $\mathcal{O}_K$, and unique ideal factorization lets us write

$$p\mathcal{O}_K = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_g^{e_g}.$$

The next phases will study the integers e_i , residue-field degrees, and ultimately the Frobenius elements that encode this splitting. Without Dedekind factorization, there would be no clean arithmetic object for Chebotarev to distribute.

Chapter 7

Synthesis: the commutative-algebra map for number theory

7.1 The seven ideas you should now connect

The goal is not to remember seven isolated definitions. The point is the chain of necessity.

1. **Ideals** generalize divisibility and survive when elements stop factoring uniquely.
2. **Prime ideals** are the atomic loci of divisibility; quotients by them are domains.
3. **Noetherianity** ensures ideals are finite objects and maximal-counterexample arguments terminate.
4. **Localization** focuses on one prime by making everything else invertible.
5. **Integrality** identifies elements that remain finite over the base arithmetic ring.
6. **DVRs** describe the local arithmetic at a single prime by one exponent.
7. **Dedekind domains** glue these local DVR pictures into global unique factorization of ideals.

The conceptual diagram is

$$\begin{array}{ccc} A & \xrightarrow{\text{localize}} & A_{\mathfrak{p}} \\ \text{integral closure} \downarrow & & \downarrow \text{Dedekind} \Rightarrow \text{DVR} \\ \mathcal{O}_K & \xrightarrow{\text{at } \mathfrak{p}} & (\mathcal{O}_K)_{\mathfrak{p}} \end{array}$$

and the arithmetic consequence is

$$\boxed{\text{global ideal}} \quad \leftrightarrow \quad \boxed{\text{one integer exponent } v_{\mathfrak{p}} \text{ at each prime } \mathfrak{p}}.$$

7.2 Proof dependency graph

You should be able to reconstruct this dependency chain from memory:

Noetherian $\implies$ finite ideals and ACC,
integral element $\iff$ finite module after adjoining it,
Dedekind $A \implies A_{\mathfrak{p}}$ is a DVR,
 $A_{\mathfrak{p}}$ DVR for each $\mathfrak{p} \implies$ nonzero ideals are locally principal,
locally principal + finite generation $\implies$ invertible ideals,
invertible ideals + Noetherian $\implies$ existence of prime ideal factorization,
prime ideal property + cancellation $\implies$ uniqueness.

Mastery checkpoint

Before doing the exercise bank, try to explain the proof of unique ideal factorization aloud without symbols. If you can say why each hypothesis of a Dedekind domain enters, you have grasped the architecture even if some technical details still need repair.

Part II

Practice problems

Chapter 8

Exercises: ideals and quotients

Attempt these before reading the solutions. Unless stated otherwise, all rings are commutative with identity.

I1. Ideals in $\mathbb{Z}$. Prove that every ideal of $\mathbb{Z}$ has the form (n) for a unique integer $n \geq 0$.

I2. Prime and maximal ideals of $\mathbb{Z}$. Classify all prime ideals and all maximal ideals of $\mathbb{Z}$.

I3. A quotient calculation. Construct an explicit isomorphism

$$\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}.$$

Identify the theorem that proves the kernel.

I4. Prime but not maximal. Show that (x) is a prime ideal of $k[x, y]$ but not a maximal ideal.

I5. Maximal ideal in two variables. Show that $(x - a, y - b)$ is maximal in $k[x, y]$ for every $a, b \in k$.

I6. Sum, intersection, and product. In $\mathbb{Z}$, compute

$$(12) + (18), \quad (12) \cap (18), \quad (12)(18).$$

Generalize your answers to $(m) + (n)$ and $(m) \cap (n)$ for positive integers m, n .

I7. Comaximal ideals. Prove that two distinct maximal ideals of a ring are comaximal.

I8. CRT computation. Find the unique class $x \pmod{280}$ satisfying

$$x \equiv 3 \pmod{8}, \quad x \equiv 4 \pmod{5}, \quad x \equiv 6 \pmod{7}.$$

Explain why uniqueness modulo 280 follows from the ring-theoretic CRT.

I9. Nilpotents and prime ideals. Prove that every nilpotent element lies in every prime ideal. Deduce that the nilradical

$$\sqrt{(0)} = \{a \in A : a^n = 0 \text{ for some } n \geq 1\}$$

is contained in the intersection of all prime ideals.

I10. Quotient correspondence. Determine all ideals of $\mathbb{Z}/12\mathbb{Z}$ and identify which are prime and which are maximal.

I11. Product rings. Determine all prime ideals of $A \times B$ in terms of prime ideals of A and B .

I12. Universal-property proof. Let $f : A \rightarrow B$ be a homomorphism. Prove the first isomorphism theorem

$$A/\ker f \cong \operatorname{im} f$$

using the universal property of the quotient.

Chapter 9

Exercises: Noetherianity

N1. ACC in practice. Show directly that every ascending chain of ideals in $\mathbb{Z}$ stabilizes.

N2. A non-Noetherian ring. Let $A = k[x_1, x_2, \dots]$. Prove that the ideal

$$I = (x_1, x_2, \dots)$$

is not finitely generated.

N3. Quotients preserve Noetherianity. Give a proof that if A is Noetherian, then A/I is Noetherian, using ascending chains rather than generators.

N4. Finite products. Prove that $A \times B$ is Noetherian if and only if both A and B are Noetherian.

N5. Finite module extension. Let A be Noetherian and let $B = A[b_1, \dots, b_r]$, where each b_i is integral over the ring generated by the previous ones. Prove that B is Noetherian.

N6. Polynomial quotients. Prove that

$$k[x_1, \dots, x_n]/I$$

is Noetherian for every ideal I .

N7. Why the basis theorem needs finite variables. Explain precisely why the chain

$$(x_1) \subsetneq (x_1, x_2) \subsetneq \dots$$

shows that Hilbert's basis theorem cannot simply be iterated infinitely many times.

N8. Maximal counterexample method. Let A be Noetherian and suppose a property P of ideals has the feature that whenever I fails P , there exists a strictly larger ideal $J \supsetneq I$ that also fails P . Prove that every ideal has P .

N9. Submodules of finite free modules. Work out the induction step in the theorem that submodules of A^n are finitely generated over a Noetherian ring, explicitly showing how generators are assembled.

N10. A finite $\mathbb{Z}$ -algebra. Show that $\mathbb{Z}[\sqrt{d}]$ is a finitely generated $\mathbb{Z}$ -module and deduce that it is Noetherian.

Chapter 10

Exercises: localization

L1. Equality of fractions. In the localization of a domain A at a multiplicative set S , prove that

$$\frac{a}{s} = \frac{b}{t} \iff at = bs.$$

Why does this equivalence need modification when zero divisors are allowed?

L2. Units in a localization. Let A be a domain. Characterize when $a/s \in S^{-1}A$ is a unit.

L3. Compute $\mathbb{Z}[1/6]$. Describe exactly which rational numbers belong to $\mathbb{Z}[1/6]$. Determine its units.

L4. Localization at (p) . Prove

$$\mathbb{Z}_{(p)} = \{a/b \in \mathbb{Q} : p \nmid b\}.$$

Determine its unique maximal ideal and its units.

L5. Surviving primes. Determine the prime ideals of $\mathbb{Z}[1/6]$.

L6. A polynomial example. Let $A = k[x, y]$ and $\mathfrak{p} = (x)$. Which elements of A become units in $A_{\mathfrak{p}}$? Is y a unit? Is $x + y$ a unit? Is x a unit?

L7. Localizing a quotient. Prove directly that

$$S^{-1}(A/I) \cong S^{-1}A/S^{-1}I,$$

where on the left one uses the image of S in A/I .

L8. Vanishing after localization. Let M be an A -module. Prove that $m/1 = 0$ in $S^{-1}M$ if and only if there exists $s \in S$ such that $sm = 0$.

L9. Detecting zero locally. Let M be an A -module. Prove that $M = 0$ if and only if $M_{\mathfrak{m}} = 0$ for every maximal ideal $\mathfrak{m}$.

L10. Equality of ideals is local. Let I, J be ideals of A . Prove that $I = J$ if and only if $I_{\mathfrak{m}} = J_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$.

L11. Localization preserves domains and fraction fields. If A is a domain and $0 \notin S$, prove $S^{-1}A$ is a domain and identify its fraction field with $\text{Frac}(A)$.

L12. Local rings. Prove that a ring A is local if and only if the set of nonunits of A is an ideal. In that case the nonunits form the unique maximal ideal.

Chapter 11

Exercises: integral dependence

G1. Integral rational numbers. Prove directly that the only rational numbers integral over $\mathbb{Z}$ are the integers.

G2. Quadratic integrality. Let d be squarefree. Determine for which integers a, b the element

$$\alpha = \frac{a + b\sqrt{d}}{2}$$

has trace and norm in $\mathbb{Z}$. Explain why this gives a useful necessary condition for integrality.

G3. Finite module from a monic equation. Suppose $b^3 + ab^2 + cb + d = 0$ with $a, c, d \in A$. Show explicitly that every element of $A[b]$ is an A -linear combination of $1, b, b^2$.

G4. Nonmonic warning. Show that $1/2$ satisfies a polynomial over $\mathbb{Z}$ but is not integral over $\mathbb{Z}$. Explain exactly why “algebraic over a ring” without monicity is too weak.

G5. Sum and product integrality. Give a proof, using the finite-module criterion rather than elimination theory, that if x, y are integral over A , then $x - y$ and xy are integral over A .

G6. Transitivity in a concrete tower. Let $A \subseteq B \subseteq C$, let $B = A[u]$ with u integral over A , and let v be integral over B . Prove directly that $A[u, v]$ is a finite A -module.

G7. UFDs are integrally closed. Reconstruct the proof that every UFD is integrally closed, identifying exactly where unique factorization is used.

G8. An integral closure example. Prove that $k[t]$ is integrally closed in $k(t)$.

G9. Localization and integrality. If B is integral over A , prove every element $b/s \in S^{-1}B$ is integral over $S^{-1}A$ by writing an explicit monic polynomial satisfied by b/s from one satisfied by b .

G10. Idempotents are integral. Show that every idempotent $e^2 = e$ is integral over every subring containing 1.

G11. Integral elements over a field. Show that if $K \subseteq L$ are fields, then “integral over K ” is equivalent to “algebraic over K ”.

G12. Finite algebra implies integral. Let B be a ring containing A . If B is finitely generated as an A -module, prove every $b \in B$ is integral over A .

Chapter 12

Exercises: DVRs and Dedekind domains

D1. Ideals of $\mathbb{Z}_{(p)}$. Prove every nonzero ideal of $\mathbb{Z}_{(p)}$ is (p^n) for a unique $n \geq 0$.

D2. Units in a DVR. Let $(A, \mathfrak{m})$ be a DVR with valuation v . Prove

$$A^\times = \{x \in K^\times : v(x) = 0\}, \quad \mathfrak{m} = \{x : v(x) > 0\}.$$

D3. Divisibility becomes order. In a DVR, prove

$$a \mid b \iff v(a) \leq v(b)$$

for nonzero $a, b \in A$.

D4. Fractional ideals of a DVR. Prove every nonzero fractional ideal of a DVR is (π^n) for a unique $n \in \mathbb{Z}$.

D5. PIDs are Dedekind. Give a complete proof that every PID is a Dedekind domain.

D6. Localization of a Dedekind domain. Let A be Dedekind and $0 \neq \mathfrak{p}$ prime. Explain in detail why the prime ideals of $A_{\mathfrak{p}}$ are only (0) and $\mathfrak{p}A_{\mathfrak{p}}$.

D7. Cancellation of ideals. If I is an invertible fractional ideal and $IJ = IK$, prove $J = K$.

D8. Product inside a prime. If $IJ \subseteq \mathfrak{p}$ and $\mathfrak{p}$ is prime, prove $I \subseteq \mathfrak{p}$ or $J \subseteq \mathfrak{p}$.

D9. Unique factorization reconstruction. Without quoting the theorem, prove existence and uniqueness of prime ideal factorization in a Dedekind domain from: (i) Noetherianity, (ii) every nonzero ideal is invertible, and (iii) every nonzero prime is maximal.

D10. Ideal exponents. Let

$$I = \mathfrak{p}^3 \mathfrak{q}^2, \quad J = \mathfrak{p} \mathfrak{q}^5 \mathfrak{r}^2$$

in a Dedekind domain. Compute IJ , $I \cap J$, $I + J$, and the fractional ideal $I/J := IJ^{-1}$ in prime-power notation. Justify the exponent rules for $I \cap J$ and $I + J$ by localizing at each prime.

D11. Failure of element factorization in $\mathbb{Z}[\sqrt{-5}]$. Prove that the units of $\mathbb{Z}[\sqrt{-5}]$ are ± 1 , and prove that $2, 3, 1 + \sqrt{-5}, 1 - \sqrt{-5}$ are irreducible. Conclude that

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$$

exhibits failure of unique factorization of elements.

D12. The ideal $\mathfrak{p}_2$. In $A = \mathbb{Z}[\sqrt{-5}]$, let $\mathfrak{p}_2 = (2, 1 + \sqrt{-5})$. Prove $A/\mathfrak{p}_2 \cong \mathbb{F}_2$ and verify directly that $\mathfrak{p}_2^2 = (2)$.

D13. Prime ideals above 3. Let

$$\mathfrak{p}_3 = (3, 1 + \sqrt{-5}), \quad \bar{\mathfrak{p}}_3 = (3, 1 - \sqrt{-5}).$$

Prove that both are maximal and that

$$(3) = \mathfrak{p}_3 \bar{\mathfrak{p}}_3.$$

D14. Reconcile the two factorizations of 6. Verify

$$(1 + \sqrt{-5}) = \mathfrak{p}_2 \mathfrak{p}_3, \quad (1 - \sqrt{-5}) = \mathfrak{p}_2 \bar{\mathfrak{p}}_3,$$

and show that both element factorizations of 6 produce the same factorization of the principal ideal (6) .

Chapter 13

Synthesis and proof reconstruction

S1. Explain localization. Give a five-minute explanation, without formulas at first, of why $A_{\mathfrak{p}}$ should be thought of as “zooming in at $\mathfrak{p}$.” Then make the explanation precise by identifying its units and prime ideals.

S2. Explain integrality. Explain why monicity forces finiteness and why this makes integral elements the correct analogue of algebraic integers.

S3. Every Dedekind hypothesis matters. In the proof that $A_{\mathfrak{p}}$ is a DVR, identify where Noetherianity, integral closedness, and dimension one are each used.

S4. Local-to-global ideal factorization. Starting with the statement “ $A_{\mathfrak{p}}$ is a DVR for every nonzero prime $\mathfrak{p}$,” explain the route to unique global ideal factorization.

S5. Element versus ideal divisibility. In $\mathbb{Z}$, relate the exponent $v_p(n)$ to the ideal factorization of (n) . Then state the Dedekind-domain analogue.

S6. Bridge to algebraic number theory. Suppose you are later told that the ring of integers $\mathcal{O}_K$ of a number field is a finite $\mathbb{Z}$ -module and integrally closed. Explain what additional fact you need to show it is Dedekind and why that conclusion immediately gives prime ideal factorization.

S7. Proof dependency map. Reconstruct, from memory, a dependency graph for the theorem “every nonzero ideal of a Dedekind domain factors uniquely into prime ideals,” listing at least six intermediate results from this phase.

S8. New example. Let $A = k[t]$ and $f = t^2(t-1)^3(t^2+1)$ over a field k where t^2+1 is irreducible. Factor the principal ideal (f) into prime ideals. Localize at $(t-1)$ and determine the localized ideal $(f)A_{(t-1)}$.

Chapter 14

Phase 1 final readiness examination

Take this closed-book after completing the exercise bank. Allow about three hours. Full solutions and a marking guide appear in Part III.

- R1.** State and prove the quotient criterion for prime ideals.
- R2.** Prove the equivalence between the ascending-chain condition and finite generation of all ideals.
- R3.** Construct $S^{-1}A$ and state/prove its universal property. Then explain why $A_{\mathfrak{p}}$ is local.
- R4.** Prove the finite-module criterion for an integral element, including the determinant trick.
- R5.** Prove that a UFD is integrally closed.
- R6.** Let $(A, \mathfrak{m})$ be a Noetherian local domain with principal maximal ideal. Prove it is a DVR.
- R7.** State the definition of a Dedekind domain and prove that $A_{\mathfrak{p}}$ is a DVR for every nonzero prime $\mathfrak{p}$.
- R8.** Prove every nonzero ideal in a Dedekind domain is invertible, using localization.
- R9.** Prove unique factorization of nonzero ideals in a Dedekind domain.
- R10.** In $\mathbb{Z}[\sqrt{-5}]$, explain rigorously how ideal factorization repairs the two incompatible element factorizations of 6.
- R11.** Explain, in one coherent paragraph, the chain

$$\text{finite } \mathbb{Z}\text{-module} \rightarrow \text{Noetherian} \rightarrow \text{Dedekind} \rightarrow \text{prime ideal factorization}$$

for rings of integers, clearly stating what extra facts are needed at each arrow.

- R12.** Give a ten-minute oral-style explanation of why all of this algebra is a prerequisite for the later definition of Frobenius and statement of Chebotarev.

Part III

Complete solutions

Chapter 15

Solutions: ideals and quotients

Solution I1

Let $I \triangleleft \mathbb{Z}$. If $I = (0)$, take $n = 0$. Otherwise I contains positive integers; by well-ordering let n be the least positive element of I . For any $a \in I$, divide $a = qn + r$ with $0 \leq r < n$. Since $r = a - qn \in I$, minimality forces $r = 0$. Hence every $a \in I$ is divisible by n , so $I \subseteq (n)$. The reverse inclusion follows from $n \in I$. Thus $I = (n)$. If $(m) = (n)$ with $m, n \geq 0$, then $m \mid n$ and $n \mid m$, hence $m = n$.

Solution I2

By I1, every ideal is (n) . The zero ideal is prime because $\mathbb{Z}$ is a domain, but it is not maximal because $(0) \subsetneq (2) \subsetneq \mathbb{Z}$. For $n > 0$, the quotient criterion gives

$$(n) \text{ prime} \iff \mathbb{Z}/n\mathbb{Z} \text{ is a domain} \iff n \text{ is prime.}$$

The same quotient is a field exactly when n is prime. Hence prime ideals are (0) and (p) for rational primes p ; maximal ideals are exactly (p) .

Solution I3

Define evaluation at i :

$$\phi : \mathbb{R}[x] \rightarrow \mathbb{C}, \quad f \mapsto f(i).$$

It is surjective because $a + bi$ is the image of $a + bx$. Since i is a root of $x^2 + 1$, the ideal $(x^2 + 1)$ lies in the kernel. Conversely, divide any f by $x^2 + 1$:

$$f = q(x)(x^2 + 1) + (ax + b).$$

If $f(i) = 0$, then $ai + b = 0$, forcing $a = b = 0$. Thus $\ker \phi = (x^2 + 1)$. The first isomorphism theorem gives the desired isomorphism.

Solution I4

We have

$$k[x, y]/(x) \cong k[y],$$

which is an integral domain, so (x) is prime. It is not maximal because $k[y]$ is not a field; for example (y) is a nonzero proper ideal. Equivalently,

$$(x) \subsetneq (x, y) \subsetneq k[x, y].$$

Solution I5

Evaluation at (a, b) gives a surjective homomorphism

$$\phi : k[x, y] \rightarrow k, \quad f \mapsto f(a, b).$$

The kernel is $(x - a, y - b)$: subtract $f(a, y)$ from $f(x, y)$ to see divisibility by $x - a$, then subtract $f(a, b)$ from $f(a, y)$ to see divisibility by $y - b$. Therefore

$$k[x, y]/(x - a, y - b) \cong k,$$

a field, so the ideal is maximal.

Solution I6

In $\mathbb{Z}$,

$$(m) + (n) = (\gcd(m, n)), \quad (m) \cap (n) = (\text{lcm}(m, n)), \quad (m)(n) = (mn).$$

The sum consists of all integer linear combinations $am + bn$, whose positive generator is the gcd. The intersection consists of integers divisible by both m and n , hence multiples of the lcm. Therefore

$$(12) + (18) = (6), \quad (12) \cap (18) = (36), \quad (12)(18) = (216).$$

Solution I7

Let $\mathfrak{m} \neq \mathfrak{n}$ be maximal. Since $\mathfrak{m} \not\subseteq \mathfrak{n}$, choose $x \in \mathfrak{m} \setminus \mathfrak{n}$. Then $\mathfrak{n} + (x)$ strictly contains $\mathfrak{n}$, so maximality of $\mathfrak{n}$ gives $\mathfrak{n} + (x) = A$. Because $(x) \subseteq \mathfrak{m}$, we get

$$A = \mathfrak{n} + (x) \subseteq \mathfrak{n} + \mathfrak{m} \subseteq A.$$

Thus $\mathfrak{m} + \mathfrak{n} = A$.

Solution I8

First combine the mod 5 and mod 7 conditions. Numbers congruent to 4 mod 5 are 4, 9, 14, 19, 24, 29, 34 (mod 35); the one congruent to 6 mod 7 is 34. Thus $x \equiv 34 \pmod{35}$. Write $x = 34 + 35k$. Modulo 8,

$$34 + 35k \equiv 2 + 3k \equiv 3,$$

so $3k \equiv 1 \pmod{8}$, hence $k \equiv 3 \pmod{8}$. Therefore

$$x \equiv 34 + 35 \cdot 3 = 139 \pmod{280}.$$

The ideals $(8), (5), (7)$ are pairwise comaximal, so CRT gives an isomorphism

$$\mathbb{Z}/(280) \cong \mathbb{Z}/(8) \times \mathbb{Z}/(5) \times \mathbb{Z}/(7),$$

which guarantees exactly one class modulo 280 for each triple of residue classes.

Solution I9

If $a^n = 0 \in \mathfrak{p}$ and $\mathfrak{p}$ is prime, then from $a^n = a \cdot a^{n-1} \in \mathfrak{p}$ we get $a \in \mathfrak{p}$ or $a^{n-1} \in \mathfrak{p}$. Repeating yields $a \in \mathfrak{p}$. Hence every nilpotent belongs to every prime ideal, so

$$\sqrt{(0)} \subseteq \bigcap_{\mathfrak{p} \in \text{Spec } A} \mathfrak{p}.$$

(The reverse inclusion is also true, but proving it requires an additional maximality argument not needed here.)

Solution I10

Ideals of $\mathbb{Z}/12\mathbb{Z}$ correspond to ideals (d) of $\mathbb{Z}$ containing (12) , equivalently positive divisors $d \mid 12$. They are

$$(0), (2), (3), (4), (6), (1) = \mathbb{Z}/12\mathbb{Z},$$

where notation now means the ideal generated by the residue class. The quotient by $(d)/(12)$ is isomorphic to $\mathbb{Z}/d\mathbb{Z}$ in the correspondence formulation. Prime/maximal ideals correspond to primes d dividing 12, namely (2) and (3) . Both are maximal because the quotients are $\mathbb{F}_2$ and $\mathbb{F}_3$.

Solution I11

Let P be prime in $A \times B$. Since

$$(1, 0)(0, 1) = (0, 0) \in P,$$

primality implies $(1, 0) \in P$ or $(0, 1) \in P$. If $(0, 1) \in P$, then $0 \times B \subseteq P$, and

$$P = \mathfrak{p} \times B$$

where $\mathfrak{p} = \{a : (a, 0) \in P\}$ is prime in A . Similarly the other case gives $A \times \mathfrak{q}$ with $\mathfrak{q}$ prime in B . Conversely,

$$(A \times B)/(\mathfrak{p} \times B) \cong A/\mathfrak{p}$$

is a domain exactly when $\mathfrak{p}$ is prime; similarly for $A \times \mathfrak{q}$.

Solution I12

Because $\ker f \subseteq \ker f$, the quotient universal property gives a unique map

$$\bar{f} : A/\ker f \rightarrow B, \quad \bar{f}(a + \ker f) = f(a).$$

Its image is exactly $\operatorname{im} f$. It is injective because

$$\bar{f}(a + \ker f) = 0 \iff f(a) = 0 \iff a \in \ker f \iff a + \ker f = 0.$$

Hence it is an isomorphism onto $\operatorname{im} f$.

Chapter 16

Solutions: Noetherianity

Solution N1

Every ideal is (n) . An inclusion $(a) \subseteq (b)$ means $b \mid a$. Thus an ascending chain

$$(n_1) \subseteq (n_2) \subseteq \cdots$$

gives a descending sequence of nonnegative integers $n_1 \geq n_2 \geq \cdots$ in the divisibility sense, and whenever inclusion is strict we have $n_{j+1} < n_j$ for positive generators. A strictly decreasing sequence of nonnegative integers cannot be infinite, so the chain stabilizes.

Solution N2

Suppose I were generated by finitely many polynomials $f_1, \dots, f_r$. Only finitely many variables occur among these polynomials; say all lie in $k[x_1, \dots, x_N]$. Then every combination

$$\sum g_i f_i$$

belongs to the ideal generated by $x_1, \dots, x_N$ and vanishes under the homomorphism setting $x_1 = \cdots = x_N = 0$. But $x_{N+1} \in I$ maps to $x_{N+1} \neq 0$, contradiction. Hence I is not finitely generated.

Solution N3

Let

$$J_1/I \subseteq J_2/I \subseteq \cdots$$

be an ascending chain in A/I . By ideal correspondence this comes from

$$J_1 \subseteq J_2 \subseteq \cdots$$

in A , with every J_i containing I . Since A is Noetherian, the latter chain stabilizes, hence so does the quotient chain.

Solution N4

If $A \times B$ is Noetherian, its quotients by $0 \times B$ and $A \times 0$ are A and B , so both are Noetherian. Conversely, every ideal of $A \times B$ has the form $I \times J$: if (a, b) lies in an ideal K , then multiplying by $(1, 0)$ and $(0, 1)$ shows $(a, 0), (0, b) \in K$. Thus $K = I \times J$. If $I = (a_1, \dots, a_r)$ and $J = (b_1, \dots, b_s)$, then K is generated by $(a_i, 0)$ and $(0, b_j)$. Hence $A \times B$ is Noetherian.

Solution N5

Set $A_0 = A$ and $A_i = A_{i-1}[b_i]$. Since b_i is integral over A_{i-1} , the finite-module criterion gives that A_i is finite as an A_{i-1} -module. Inductively, a finite module over a finite module is finite, so $B = A_r$ is a finite A -module. Because A is Noetherian, the finite-module theorem implies B is Noetherian.

Solution N6

A field k is Noetherian because its only ideals are 0 and k . Hilbert's basis theorem, iterated n times, gives that $k[x_1, \dots, x_n]$ is Noetherian. Quotients of Noetherian rings are Noetherian, so the quotient by any ideal I is Noetherian.

Solution N7

Hilbert's basis theorem says that one polynomial extension of a Noetherian ring is Noetherian. Any finite iteration therefore remains Noetherian. But no finite stage contains all variables in $k[x_1, x_2, \dots]$. The displayed infinite chain is strictly increasing because $x_{n+1} \notin (x_1, \dots, x_n)$, as seen by setting $x_1 = \dots = x_n = 0$. Thus the direct union of the finite-variable polynomial rings is not Noetherian. No theorem says Noetherianity is preserved under an infinite directed union.

Solution N8

Assume some ideal fails P . Let $\mathcal{F}$ be the nonempty collection of ideals failing P . Noetherianity implies every nonempty family of ideals has a maximal element, so choose maximal $I \in \mathcal{F}$. By hypothesis there exists $J \supsetneq I$ also failing P , contradicting maximality. Therefore no counterexample exists.

Solution N9

Let $N \subseteq A^n$ and project to the last coordinate. Its image is an ideal $I = (a_1, \dots, a_r)$. Choose $v_i \in N$ with last coordinate a_i . The kernel of the projection restricted to N is

$$K = N \cap (A^{n-1} \times 0),$$

which is finitely generated by induction, say $w_1, \dots, w_s$. For $v \in N$ with last coordinate a , write $a = \sum c_i a_i$. Then

$$v - \sum c_i v_i \in K,$$

so it is an A -linear combination of the w_j . Therefore the finite set $\{v_i, w_j\}$ generates N .

Solution N10

Every polynomial in $\sqrt{d}$ reduces using $(\sqrt{d})^2 = d$ to

$$a + b\sqrt{d}, \quad a, b \in \mathbb{Z}.$$

Thus $\mathbb{Z}[\sqrt{d}]$ is generated as a $\mathbb{Z}$ -module by 1 and $\sqrt{d}$. Since $\mathbb{Z}$ is Noetherian and the ring is finite as a $\mathbb{Z}$ -module, it is Noetherian.

Chapter 17

Solutions: localization

Solution L1

For a domain with $0 \notin S$, the localization relation says $a/s = b/t$ if there exists $u \in S$ with

$$u(at - bs) = 0.$$

Since $u \neq 0$ and the ring has no zero divisors, this is equivalent to $at - bs = 0$. With zero divisors, a nonzero u may annihilate $at - bs$, so equality can hold even when $at \neq bs$; the extra multiplier is needed to make the fraction relation transitive and compatible with the intended universal property.

Solution L2

The fraction a/s is a unit exactly when there exist $b \in A$ and $t \in S$ with

$$\frac{a}{s} \frac{b}{t} = 1,$$

which in a domain is equivalent to $ab = st$. Since $st \in S$, this says precisely that a divides some element of S . Thus

$$a/s \in (S^{-1}A)^\times \iff \exists b \in A \text{ with } ab \in S.$$

If S is saturated, this reduces to $a \in S$.

Solution L3

Every element is $a/6^n$. After cancellation, its denominator has no prime factors except 2 and 3. Conversely, every rational number whose reduced denominator is $2^r 3^s$ can be rewritten over a sufficiently large power of 6. Hence

$$\mathbb{Z}[1/6] = \left\{ \frac{a}{b} \in \mathbb{Q} : \text{all prime divisors of } b \text{ are } 2 \text{ or } 3 \right\}.$$

A rational number is a unit exactly when both it and its reciprocal lie in the ring, so its

numerator and denominator may involve only 2 and 3. Therefore

$$\mathbb{Z}[1/6]^\times = \{\pm 2^m 3^n : m, n \in \mathbb{Z}\}.$$

Solution L4

Localizing $\mathbb{Z}$ at $S = \mathbb{Z} \setminus (p)$ allows exactly denominators not divisible by p , giving

$$\mathbb{Z}_{(p)} = \{a/b \in \mathbb{Q} : p \nmid b\}.$$

A fraction a/b is a unit iff its numerator is also not divisible by p , because then b/a lies in the ring. Therefore the nonunits are precisely fractions with $p \mid a$, which form the unique maximal ideal $p\mathbb{Z}_{(p)}$.

Solution L5

Prime ideals of the localization correspond to primes of $\mathbb{Z}$ disjoint from $S = \{6^n : n \geq 0\}$. The prime ideals of $\mathbb{Z}$ are (0) and (p) . The ideal (p) meets S exactly when $p \mid 6$, i.e. $p = 2$ or 3 . Hence the primes are

$$(0), \quad (p)\mathbb{Z}[1/6] \quad (p \text{ prime}, p \neq 2, 3).$$

Solution L6

In $A_{(x)}$, every polynomial not in (x) becomes a unit. A polynomial lies in (x) exactly when every term is divisible by x , equivalently its specialization at $x = 0$ is zero. Thus $y \notin (x)$ and is a unit; $x + y \notin (x)$ and is a unit; $x \in (x)$ and is not a unit. More generally, all nonzero polynomials in $k[y]$ become units.

Solution L7

Define

$$\Phi : S^{-1}A \rightarrow S^{-1}(A/I), \quad a/s \mapsto (a + I)/(s + I).$$

It is surjective. Its kernel consists of a/s for which $(a + I)/(s + I) = 0$. By the localization zero criterion, this means there exists $t \in S$ such that $ta \in I$, which is exactly the condition $a/s \in S^{-1}I$. Thus

$$\ker \Phi = S^{-1}I.$$

The first isomorphism theorem gives

$$S^{-1}A/S^{-1}I \cong S^{-1}(A/I).$$

Solution L8

By definition of equality in the localized module,

$$\frac{m}{1} = \frac{0}{1}$$

iff there exists $s \in S$ such that

$$s(1 \cdot m - 1 \cdot 0) = sm = 0.$$

This criterion is one of the most useful ways to think about localization: an element disappears exactly when some denominator you have allowed kills it.

Solution L9

If $M = 0$, all localizations are zero. Conversely suppose $M \neq 0$ and choose $0 \neq m \in M$. Its annihilator

$$I = \{a \in A : am = 0\}$$

is a proper ideal, hence lies in some maximal ideal $\mathfrak{m}$. If $m/1 = 0$ in $M_{\mathfrak{m}}$, L8 gives $s \notin \mathfrak{m}$ with $sm = 0$, so $s \in I \subseteq \mathfrak{m}$, contradiction. Therefore $M_{\mathfrak{m}} \neq 0$.

Solution L10

If $I = J$, localizations are equal. Conversely suppose $I_{\mathfrak{m}} = J_{\mathfrak{m}}$ for all maximal $\mathfrak{m}$. Apply L9 to the module $(I + J)/I$: its localization is zero because

$$((I + J)/I)_{\mathfrak{m}} \cong (I_{\mathfrak{m}} + J_{\mathfrak{m}})/I_{\mathfrak{m}} = 0.$$

Hence $(I + J)/I = 0$, so $J \subseteq I$. Reversing the roles gives $I \subseteq J$.

Solution L11

If $(a/s)(b/t) = 0$ in the localization, then some $u \in S$ satisfies $uab = 0$. In a domain $u \neq 0$, so $a = 0$ or $b = 0$, hence one fraction is zero. Thus $S^{-1}A$ is a domain. It lies naturally inside $\text{Frac}(A)$ by $a/s \mapsto a/s$. Its own fraction field contains every a/b with $a, b \in A$, $b \neq 0$, because $a/b = (a/1)/(b/1)$. Hence

$$\text{Frac}(S^{-1}A) = \text{Frac}(A).$$

Solution L12

If A is local with maximal ideal $\mathfrak{m}$, every element outside $\mathfrak{m}$ is a unit: if $a \notin \mathfrak{m}$, the ideal (a) is not contained in the unique maximal ideal, so $(a) = A$. Hence the nonunits are exactly $\mathfrak{m}$, an ideal.

Conversely suppose the nonunits form an ideal N . It is proper because 1 is a unit. Every proper ideal contains no units, hence lies in N . Thus N contains every proper ideal and is therefore the unique maximal ideal. So A is local.

Chapter 18

Solutions: integral dependence

Solution G1

Let $r = a/b \in \mathbb{Q}$ be in lowest terms with $b > 0$, and suppose r is integral over $\mathbb{Z}$. Then

$$r^n + c_{n-1}r^{n-1} + \cdots + c_0 = 0$$

with $c_i \in \mathbb{Z}$. Multiplying by b^n gives

$$a^n = -b(c_{n-1}a^{n-1} + c_{n-2}a^{n-2}b + \cdots + c_0b^{n-1}).$$

Hence $b \mid a^n$. Since $\gcd(a, b) = 1$, this forces $b = 1$. Thus $r \in \mathbb{Z}$. Conversely every integer is integral because it satisfies $T - r = 0$.

Solution G2

The conjugate of

$$\alpha = \frac{a + b\sqrt{d}}{2}$$

is $(a - b\sqrt{d})/2$. Therefore

$$\mathrm{Tr}(\alpha) = a, \quad N(\alpha) = \frac{a^2 - b^2d}{4}.$$

The trace is automatically integral. The norm is integral exactly when

$$a^2 \equiv b^2d \pmod{4}.$$

Squares mod 4 are 0 and 1. If b is even, then the right side is 0, so a must be even. If b is odd, then $b^2 \equiv 1$, so we need $a^2 \equiv d \pmod{4}$. This is possible exactly when $d \equiv 1 \pmod{4}$ and a is odd. Thus trace and norm are both integers exactly when either a, b are both even, or $d \equiv 1 \pmod{4}$ and a, b are both odd.

For a quadratic element, integrality implies integral trace and norm because its monic minimal polynomial is $T^2 - \mathrm{Tr}(\alpha)T + N(\alpha)$. Conversely, if these two quantities are integers, that same polynomial proves integrality. Thus here the condition is actually necessary and sufficient.

Solution G3

From

$$b^3 = -ab^2 - cb - d$$

we can replace every occurrence of b^3 by a combination of lower powers. Multiplying this relation by b expresses b^4 in terms of $1, b, b^2$, and induction does the same for all b^n , $n \geq 3$. Hence every polynomial in b reduces to

$$r_0 + r_1b + r_2b^2, \quad r_i \in A.$$

Therefore $A[b]$ is generated as an A -module by $1, b, b^2$.

Solution G4

The rational number $1/2$ satisfies the polynomial

$$2T - 1 = 0$$

over $\mathbb{Z}$, but by G1 it is not integral. The problem is that the leading coefficient is not 1. From a monic equation one can solve for the highest power as an A -linear combination of lower powers; from $2b = 1$ one instead needs to divide by 2, which is precisely the new denominator that integrality is designed to forbid.

Solution G5

Because x and y are integral, $A[x]$ is finite over A and $A[x, y]$ is finite over $A[x]$, hence finite over A . The finite A -module $M = A[x, y]$ contains 1. Multiplication by $x - y$ maps M to itself, as does multiplication by xy . Applying the determinant trick to these endomorphisms and then evaluating the resulting annihilating polynomial at $1 \in M$ produces monic equations over A . Hence $x - y$ and xy are integral.

Solution G6

Suppose u satisfies a monic equation of degree m over A . Then $A[u]$ is generated over A by

$$1, u, \dots, u^{m-1}.$$

Suppose v satisfies a monic equation of degree n over $A[u]$. Then $A[u, v]$ is generated over $A[u]$ by

$$1, v, \dots, v^{n-1}.$$

Combining the generating sets, $A[u, v]$ is generated over A by the mn products

$$u^i v^j, \quad 0 \leq i < m, \quad 0 \leq j < n.$$

Thus it is a finite A -module.

Solution G7

Let $x = a/b \in \text{Frac}(A)$ be integral with a, b chosen coprime in the UFD A . A monic equation gives after clearing denominators

$$a^n = -b(\cdots),$$

so $b \mid a^n$. If an irreducible π divides b , unique factorization implies irreducibles are prime in a UFD, hence $\pi \mid a^n$ implies $\pi \mid a$. That contradicts coprimality. Therefore b has no irreducible factors and must be a unit. Hence $x \in A$. The exact use of UFD is the implication “an irreducible divisor of a^n divides a ,” equivalently that irreducibles are prime.

Solution G8

The polynomial ring $k[t]$ is a Euclidean domain with degree as Euclidean function, hence a PID and therefore a UFD. By the theorem just proved, every UFD is integrally closed in its fraction field. Since $\text{Frac}(k[t]) = k(t)$, the integral closure of $k[t]$ in $k(t)$ is $k[t]$ itself.

Solution G9

Suppose

$$b^n + a_{n-1}b^{n-1} + \cdots + a_0 = 0, \quad a_i \in A.$$

Let $y = b/s$. Since $b = sy$, substitute to get

$$s^n y^n + a_{n-1} s^{n-1} y^{n-1} + \cdots + a_0 = 0.$$

Divide by the unit s^n in $S^{-1}A$:

$$y^n + \frac{a_{n-1}}{s} y^{n-1} + \frac{a_{n-2}}{s^2} y^{n-2} + \cdots + \frac{a_0}{s^n} = 0.$$

This is monic with coefficients in $S^{-1}A$, so b/s is integral over $S^{-1}A$.

Solution G10

If $e^2 = e$, then e satisfies the monic polynomial

$$T^2 - T = 0.$$

Its coefficients are 0 and -1 , which belong to every subring containing 1. Hence e is integral over any such subring.

Solution G11

If α is integral over a field K , then it satisfies a nonzero polynomial over K , hence is algebraic. Conversely, if α is algebraic, its minimal polynomial over K can be chosen monic because its leading coefficient is a nonzero element of the field and can be divided out. Thus α is integral. Over fields, algebraic and integral therefore coincide.

Solution G12

Fix $b \in B$. Multiplication by b is an A -linear endomorphism of the finite A -module B . Choose generators $m_1, \dots, m_r$. We can write

$$bm_i = \sum_j a_{ij} m_j.$$

As in the determinant trick,

$$(bI - C)\mathbf{m} = 0.$$

Multiplying by the adjugate yields

$$\det(bI - C)m_i = 0$$

for every generator. Since those generators span B , the same element annihilates $1 \in B$. Thus $\det(bI - C) \cdot 1 = 0$, so $\det(bI - C) = 0$ in B . This is a monic polynomial equation for b over A .

Chapter 19

Solutions: DVRs and Dedekind domains

Solution D1

Take a nonzero ideal $I \subseteq \mathbb{Z}_{(p)}$. For each nonzero $x = a/b \in I$ with $p \nmid b$, write

$$x = p^{v_p(a)}u$$

where u has numerator and denominator prime to p , hence is a unit. Choose an element of I for which the exponent $n = v_p(x)$ is minimal. Then $(x) = (p^n) \subseteq I$. Every $y \in I$ has $v_p(y) \geq n$, so $y/p^n \in \mathbb{Z}_{(p)}$ and $y \in (p^n)$. Thus $I = (p^n)$. Uniqueness follows because $(p^n) = (p^m)$ would imply p^{n-m} is a unit, forcing $n = m$.

Solution D2

If $v(x) = 0$, then $v(x^{-1}) = 0$, so both x and x^{-1} lie in A ; hence x is a unit. If x is a unit, $x, x^{-1} \in A$ give $v(x), v(x^{-1}) \geq 0$, but their sum is 0, so both valuations are 0. Therefore

$$A^\times = \{x : v(x) = 0\}.$$

An element of A is a nonunit exactly when its valuation is positive, and in a local ring the nonunits are the maximal ideal. Hence

$$\mathfrak{m} = \{x : v(x) > 0\}.$$

Solution D3

If $a \mid b$, write $b = ac$ with $c \in A$. Then

$$v(b) = v(a) + v(c) \geq v(a).$$

Conversely if $v(a) \leq v(b)$, then

$$v(b/a) = v(b) - v(a) \geq 0,$$

so $b/a \in A$ and $b = a(b/a)$. Thus $a \mid b$.

Solution D4

Let I be a nonzero fractional ideal. Choose $0 \neq d \in A$ with $dI \subseteq A$. The ordinary ideal dI equals (π^m) for some $m \geq 0$, while $(d) = (\pi^r)$ for some $r \geq 0$ up to a unit. Hence

$$I = d^{-1}(dI) = (\pi^{m-r}).$$

The exponent can now be any integer. Uniqueness follows by valuation.

Solution D5

Let A be a PID.

1. It is Noetherian because every ideal has one generator.
2. Every PID is a UFD, and every UFD is integrally closed, so A is integrally closed in its fraction field.
3. Let (π) be a nonzero prime ideal. Then π is a prime element, hence irreducible. In a PID, if an ideal strictly contains (π) , it is (d) with $d \mid \pi$. Irreducibility forces d to be a unit or associate to π , so the containing ideal is A or (π) . Hence (π) is maximal.

Thus A satisfies all three Dedekind conditions.

Solution D6

Primes of $A_{\mathfrak{p}}$ correspond to primes $\mathfrak{q}$ of A with $\mathfrak{q} \subseteq \mathfrak{p}$. Since A is a domain, (0) is prime and survives. If $\mathfrak{q} \neq 0$, the Dedekind condition says $\mathfrak{q}$ is maximal. But $\mathfrak{q} \subseteq \mathfrak{p}$ and $\mathfrak{p}$ is also a proper ideal, so maximality forces $\mathfrak{q} = \mathfrak{p}$. Therefore the only prime ideals of $A_{\mathfrak{p}}$ are (0) and $\mathfrak{p}A_{\mathfrak{p}}$.

Solution D7

Multiply the equality $IJ = IK$ by I^{-1} :

$$I^{-1}IJ = I^{-1}IK.$$

Associativity of ideal multiplication gives

$$(I^{-1}I)J = (I^{-1}I)K.$$

Since I is invertible, $I^{-1}I = A$, so $AJ = AK$, i.e. $J = K$.

Solution D8

Suppose neither I nor J is contained in $\mathfrak{p}$. Choose $a \in I \setminus \mathfrak{p}$ and $b \in J \setminus \mathfrak{p}$. Then $ab \in IJ \subseteq \mathfrak{p}$. Since $\mathfrak{p}$ is prime, $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$, contradiction. Therefore $I \subseteq \mathfrak{p}$ or $J \subseteq \mathfrak{p}$.

Solution D9

Existence. Assume some nonzero proper ideal is not a product of prime ideals. Noetherianity implies the counterexamples have a maximal member I . Choose a maximal ideal $\mathfrak{p} \supseteq I$. Since

$\mathfrak{p}$ is nonzero and all nonzero primes are maximal, it is a permissible prime factor. Set

$$J = \mathfrak{p}^{-1}I.$$

Because $I \subseteq \mathfrak{p}$, $J \subseteq A$, and $I = \mathfrak{p}J$. Also $I \subsetneq J$: if $I = J$, then $I = \mathfrak{p}I$ and cancellation by I^{-1} gives $A = \mathfrak{p}$, impossible. By maximality of I , the larger ideal J factors into primes; multiplying by $\mathfrak{p}$ factors I , contradiction.

Uniqueness. Suppose

$$\mathfrak{p}_1 \cdots \mathfrak{p}_r = \mathfrak{q}_1 \cdots \mathfrak{q}_s.$$

The right side is contained in $\mathfrak{p}_1$. Repeated use of D8 implies some $\mathfrak{q}_j \subseteq \mathfrak{p}_1$. Since both are nonzero prime ideals and therefore maximal, $\mathfrak{q}_j = \mathfrak{p}_1$. Cancel this invertible factor. Induction finishes the proof.

Solution D10

Prime exponents add under multiplication and negate under inversion, so

$$IJ = \mathfrak{p}^4 \mathfrak{q}^7 \mathfrak{r}^2, \quad IJ^{-1} = \mathfrak{p}^2 \mathfrak{q}^{-3} \mathfrak{r}^{-2}.$$

For sums and intersections, localize at a prime ℓ . In the DVR A_ℓ , ideals are linearly ordered:

$$(\pi^a) + (\pi^b) = (\pi^{\min(a,b)}), \quad (\pi^a) \cap (\pi^b) = (\pi^{\max(a,b)}).$$

Since equality of ideals can be checked locally, globally

$$v_\ell(I + J) = \min(v_\ell(I), v_\ell(J)), \quad v_\ell(I \cap J) = \max(v_\ell(I), v_\ell(J)).$$

Therefore

$$I + J = \mathfrak{p}\mathfrak{q}^2, \quad I \cap J = \mathfrak{p}^3 \mathfrak{q}^5 \mathfrak{r}^2.$$

Solution D11

For $z = a + b\sqrt{-5}$ define

$$N(z) = a^2 + 5b^2.$$

The norm is multiplicative. If z is a unit, $N(z)N(z^{-1}) = 1$, so $N(z) = 1$, forcing $b = 0$ and $a = \pm 1$. Thus the units are ± 1 .

Now $N(2) = 4$. A nontrivial factorization $2 = xy$ would require norms $N(x), N(y) > 1$ multiplying to 4, hence both equal 2. But $a^2 + 5b^2 = 2$ has no integer solutions. Thus 2 is irreducible. Similarly $N(3) = 9$; a nontrivial factorization would require an element of norm 3, but $a^2 + 5b^2 = 3$ has no solution. Hence 3 is irreducible.

Finally

$$N(1 \pm \sqrt{-5}) = 6.$$

A nontrivial factorization would require norms 2 and 3, neither of which occurs. So both are irreducible. The two displayed factorizations of 6 involve irreducibles that are not associates (their norms and forms differ), proving failure of unique factorization.

Solution D12

Define

$$\phi : A \rightarrow \mathbb{F}_2, \quad a + b\sqrt{-5} \mapsto a + b \pmod{2}.$$

This is a ring homomorphism because in $\mathbb{F}_2$, the proposed image of $\sqrt{-5}$ is 1, and $1^2 = 1 \equiv -5 \pmod{2}$. It is surjective. Its kernel consists of $a + b\sqrt{-5}$ with $a + b$ even, and such an element can be written as

$$b(1 + \sqrt{-5}) + (a - b),$$

where $a - b$ is even, so the kernel is $(2, 1 + \sqrt{-5}) = \mathfrak{p}_2$. Hence $A/\mathfrak{p}_2 \cong \mathbb{F}_2$.

Now $\mathfrak{p}_2^2$ is generated by

$$4, \quad 2(1 + \sqrt{-5}), \quad (1 + \sqrt{-5})^2 = -4 + 2\sqrt{-5}.$$

All are in (2) , so $\mathfrak{p}_2^2 \subseteq (2)$. Conversely,

$$2(1 + \sqrt{-5}) - (1 + \sqrt{-5})^2 = 6,$$

so $6 \in \mathfrak{p}_2^2$, while $4 \in \mathfrak{p}_2^2$. Thus $2 = 6 - 4 \in \mathfrak{p}_2^2$, giving $(2) \subseteq \mathfrak{p}_2^2$. Hence equality.

Solution D13

For $\mathfrak{p}_3$, define

$$\phi_+ : A \rightarrow \mathbb{F}_3, \quad a + b\sqrt{-5} \mapsto a - b \pmod{3},$$

because $\sqrt{-5}$ is sent to -1 , and $(-1)^2 = 1 \equiv -5 \pmod{3}$. Its kernel is $(3, 1 + \sqrt{-5}) = \mathfrak{p}_3$, so $A/\mathfrak{p}_3 \cong \mathbb{F}_3$. Similarly sending $\sqrt{-5}$ to 1 gives kernel $\bar{\mathfrak{p}}_3$ and quotient $\mathbb{F}_3$. Both ideals are maximal.

Their product is generated by

$$9, \quad 3(1 + \sqrt{-5}), \quad 3(1 - \sqrt{-5}), \quad (1 + \sqrt{-5})(1 - \sqrt{-5}) = 6.$$

Every generator is divisible by 3, so the product lies in (3) . Conversely, $9 - 6 = 3$ belongs to the product, hence $(3) \subseteq \mathfrak{p}_3\bar{\mathfrak{p}}_3$. Thus

$$(3) = \mathfrak{p}_3\bar{\mathfrak{p}}_3.$$

Solution D14

Every generator of $\mathfrak{p}_2\mathfrak{p}_3$ is divisible by $1 + \sqrt{-5}$:

$$6 = (1 + \sqrt{-5})(1 - \sqrt{-5}),$$

and the other generators are $2(1 + \sqrt{-5})$, $3(1 + \sqrt{-5})$, and $(1 + \sqrt{-5})^2$. Thus

$$\mathfrak{p}_2\mathfrak{p}_3 \subseteq (1 + \sqrt{-5}).$$

Conversely,

$$3(1 + \sqrt{-5}) - 2(1 + \sqrt{-5}) = 1 + \sqrt{-5}$$

belongs to the product, so equality holds. Conjugating gives

$$(1 - \sqrt{-5}) = \mathfrak{p}_2\bar{\mathfrak{p}}_3.$$

Hence

$$(2)(3) = \mathfrak{p}_2^2 \mathfrak{p}_3 \bar{\mathfrak{p}}_3$$

and also

$$(1 + \sqrt{-5})(1 - \sqrt{-5}) = (\mathfrak{p}_2 \mathfrak{p}_3)(\mathfrak{p}_2 \bar{\mathfrak{p}}_3) = \mathfrak{p}_2^2 \mathfrak{p}_3 \bar{\mathfrak{p}}_3.$$

The two element factorizations therefore become the same unique ideal factorization.

Chapter 20

Solutions: synthesis

Solution S1

To “zoom in at $\mathfrak{p}$ ” means to treat everything not controlled by $\mathfrak{p}$ as harmless. Algebraically, harmless elements should become units, so we invert every element outside $\mathfrak{p}$. The resulting ring is $A_{\mathfrak{p}}$. Its units are precisely fractions a/s with $a \notin \mathfrak{p}$, while the nonunits are those with numerator in $\mathfrak{p}$; these form the unique maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$. Its prime ideals correspond to primes $\mathfrak{q} \subseteq \mathfrak{p}$. Thus all primes not lying under $\mathfrak{p}$ disappear from view.

Solution S2

If x satisfies a monic equation of degree n , then x^n is an A -linear combination of lower powers. Consequently all powers reduce to the finite span of $1, x, \dots, x^{n-1}$, so adjoining x creates only finitely much new A -module information. This is exactly what one wants from an “integer-like” algebraic element: it may live in a larger field, but it does not introduce arbitrary denominators. Over $\mathbb{Z}$, these are precisely the algebraic integers.

Solution S3

In the proof for a local domain $(A, \mathfrak{m})$:

- **Dimension one** is used to conclude that for $0 \neq x \in \mathfrak{m}$, every prime containing (x) is $\mathfrak{m}$, hence $\sqrt{(x)} = \mathfrak{m}$.
- **Noetherianity** is used first to finitely generate $\mathfrak{m}$, which turns $\sqrt{(x)} = \mathfrak{m}$ into $\mathfrak{m}^N \subseteq (x)$, and later through the determinant trick/finite-module argument. It is also what guarantees the principal-maximal-ideal process terminates when proving DVR structure.
- **Integral closedness** is used at the decisive contradiction: if multiplication by $t = y/x$ preserved $\mathfrak{m}$, the determinant trick would make t integral, so integrally closedness would force $t \in A$, contrary to construction.

All three hypotheses have distinct roles.

Solution S4

If every $A_{\mathfrak{p}}$ is a DVR, then every localized nonzero ideal $I_{\mathfrak{p}}$ is principal, in fact a power of $\mathfrak{p}A_{\mathfrak{p}}$. Because A is Noetherian, I is finitely generated, which lets inverse ideals commute with localization. Locally a principal ideal is invertible, so $(II^{-1})_{\mathfrak{p}} = A_{\mathfrak{p}}$ for every maximal $\mathfrak{p}$. Local equality then gives $II^{-1} = A$ globally. Once all nonzero ideals are invertible, Noetherianity yields a maximal counterexample to existence of prime factorization; dividing that counterexample by a containing prime produces a larger ideal and gives the contradiction. Uniqueness follows from primality plus cancellation by invertible ideals.

Solution S5

For $n \neq 0$ in $\mathbb{Z}$,

$$(n) = \prod_p (p)^{v_p(n)}.$$

Thus the ordinary p -adic exponent is exactly the exponent of the prime ideal (p) in the ideal factorization of (n) . In a Dedekind domain, for every nonzero prime ideal $\mathfrak{p}$ we similarly have a valuation $v_{\mathfrak{p}}$, and for $x \in K^{\times}$,

$$(x) = \prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}(x)},$$

with only finitely many nonzero exponents.

Solution S6

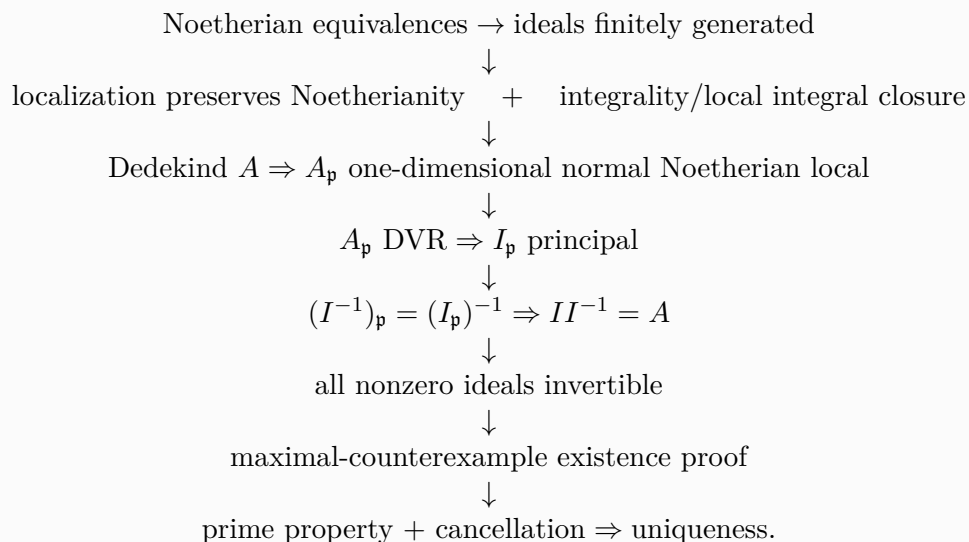
A finite $\mathbb{Z}$ -module is Noetherian because $\mathbb{Z}$ is Noetherian and submodules of finite modules are finitely generated. If $\mathcal{O}_K$ is also integrally closed, then two of the three Dedekind conditions are known. The remaining fact is that every nonzero prime ideal is maximal, equivalently that $\mathcal{O}_K$ has dimension one. Once this is established, $\mathcal{O}_K$ is Dedekind, and the unique ideal factorization theorem immediately gives

$$\mathfrak{a} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r}$$

for every nonzero ideal.

Solution S7

One valid dependency graph is:



Other graphs are acceptable if the logical dependencies are correct.

Solution S8

Since $k[t]$ is a PID, its nonzero prime ideals are generated by irreducible polynomials. Therefore

$$(f) = (t)^2(t-1)^3(t^2+1).$$

When we localize at $\mathfrak{p} = (t-1)$, both t and t^2+1 lie outside $\mathfrak{p}$ and become units. Hence their ideal factors become the unit ideal. Only the $\mathfrak{p}$ -power survives:

$$(f)A_{(t-1)} = (t-1)^3A_{(t-1)}.$$

This is localization making “all other primes” invisible.

Chapter 21

Solutions and marking guide: final readiness examination

Solution R1

For a proper ideal $I \triangleleft A$,

$$I \text{ is prime} \iff A/I \text{ is an integral domain.}$$

Indeed,

$$(a + I)(b + I) = 0 + I \iff ab \in I.$$

Thus the statement that a product of two nonzero cosets can never be zero is exactly the statement

$$ab \in I \Rightarrow a \in I \text{ or } b \in I.$$

This is the prime-ideal condition.

Solution R2

Assume every ideal is finitely generated. Given

$$I_1 \subseteq I_2 \subseteq \cdots,$$

let $I = \bigcup I_n$. The chain condition makes I an ideal. Write $I = (a_1, \dots, a_r)$. All generators occur in one sufficiently large I_N , so $I = I_N$ and the chain stabilizes.

Conversely assume ACC. For an ideal I , if it is not finitely generated choose $a_1 \in I$, then $a_2 \in I \setminus (a_1)$, and inductively $a_{n+1} \notin (a_1, \dots, a_n)$. This creates a strictly ascending chain

$$(a_1) \subsetneq (a_1, a_2) \subsetneq \cdots,$$

contradicting ACC. Hence every ideal is finitely generated.

Solution R3

Let $S \subseteq A$ be multiplicatively closed. Form symbols a/s and identify

$$(a, s) \sim (b, t)$$

when some $u \in S$ satisfies $u(at - bs) = 0$. Define addition and multiplication by the usual fraction rules. The canonical map is $i : A \rightarrow S^{-1}A$, $a \mapsto a/1$.

The universal property says: if $f : A \rightarrow B$ sends every $s \in S$ to a unit, then there is a unique map

$$\tilde{f} : S^{-1}A \rightarrow B, \quad \tilde{f}(a/s) = f(a)f(s)^{-1},$$

with $\tilde{f} \circ i = f$. Well-definedness follows from applying f to $u(at - bs) = 0$ and cancelling the unit $f(u)$. Uniqueness is forced because $a/s = (a/1)(s/1)^{-1}$.

For a prime $\mathfrak{p}$, take $S = A \setminus \mathfrak{p}$. A fraction a/s with $a \notin \mathfrak{p}$ has inverse s/a , while a fraction with $a \in \mathfrak{p}$ belongs to $\mathfrak{p}A_{\mathfrak{p}}$. Hence the nonunits are exactly $\mathfrak{p}A_{\mathfrak{p}}$, so $A_{\mathfrak{p}}$ is local with unique maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$.

Solution R4

If b satisfies a monic degree- n equation over A , then every power b^m with $m \geq n$ reduces to an A -linear combination of $1, b, \dots, b^{n-1}$. Thus $A[b]$ is finite over A .

Conversely suppose a finite A -submodule $M = (m_1, \dots, m_r)$ contains 1 and is stable under multiplication by b . Write

$$bm_i = \sum_j a_{ij}m_j.$$

Then

$$(bI - C)\mathbf{m} = 0.$$

Multiplying by the adjugate gives

$$\det(bI - C)\mathbf{m} = 0.$$

The determinant is monic of degree r in b and annihilates every element of M . Since $1 \in M$, it annihilates 1, hence vanishes as an element of the ring. Thus b satisfies a monic polynomial over A and is integral.

Solution R5

Let $x = a/b$ in the fraction field of the UFD A , with a, b coprime. If x is integral, then after clearing denominators from a monic equation we obtain $b \mid a^n$. Every irreducible divisor of b is prime in a UFD, hence would divide a , contradicting coprimality. Therefore b is a unit and $x \in A$. Thus A is integrally closed.

Solution R6

Write $\mathfrak{m} = (\pi)$. For $0 \neq x \in A$, if x is a nonunit then $x = \pi x_1$. Continue while x_i is a nonunit. The process cannot continue indefinitely, because it would yield

$$(x) \subsetneq (x_1) \subsetneq (x_2) \subsetneq \cdots,$$

contradicting Noetherianity. Hence $x = u\pi^n$ for a unit u . For $x = a/b \in K^\times$, define

$$v(x) = n_a - n_b$$

from these factorizations. Units do not affect the exponent, so v is well defined, additive under products, and satisfies the ultrametric inequality. It is surjective because $v(\pi) = 1$, and $A = \{x : v(x) \geq 0\}$. Hence A is a DVR.

Solution R7

A Dedekind domain is a Noetherian, integrally closed domain in which every nonzero prime ideal is maximal. Let $0 \neq \mathfrak{p}$ be prime. The localization $A_{\mathfrak{p}}$ remains Noetherian. Integral closedness localizes: if x is integral over $A_{\mathfrak{p}}$, clear the finitely many denominators in a monic equation so that sx is integral over A for some $s \notin \mathfrak{p}$; then $sx \in A$ and $x \in A_{\mathfrak{p}}$. Prime ideals of $A_{\mathfrak{p}}$ correspond to primes of A contained in $\mathfrak{p}$. Since every nonzero prime is maximal, these are only (0) and $\mathfrak{p}$. Thus $A_{\mathfrak{p}}$ is a one-dimensional Noetherian integrally closed local domain. The local theorem proved in the theory section shows such a ring is a DVR.

Solution R8

Let I be a nonzero ideal. Since A is Noetherian, I is finitely generated. Define

$$I^{-1} = \{x \in K : xI \subseteq A\}.$$

For each maximal $\mathfrak{p}$,

$$(I^{-1})_{\mathfrak{p}} = (I_{\mathfrak{p}})^{-1};$$

finite generation lets us clear finitely many denominators to prove the reverse inclusion. Because $A_{\mathfrak{p}}$ is a DVR, $I_{\mathfrak{p}}$ is principal and therefore invertible. Hence

$$(II^{-1})_{\mathfrak{p}} = A_{\mathfrak{p}}$$

for every maximal $\mathfrak{p}$. If II^{-1} were proper, it would be contained in some maximal $\mathfrak{p}$, making its localization proper. Contradiction. Thus $II^{-1} = A$.

Solution R9

For existence, assume counterexamples exist and choose a maximal one I by Noetherianity. Take a maximal ideal $\mathfrak{p} \supseteq I$ and set $J = \mathfrak{p}^{-1}I$. Then $J \subseteq A$, $I = \mathfrak{p}J$, and J strictly contains I ; otherwise cancellation would give $\mathfrak{p} = A$. Maximality of I implies J factors into prime ideals, so I does too, contradiction.

For uniqueness, if

$$\mathfrak{p}_1 \cdots \mathfrak{p}_r = \mathfrak{q}_1 \cdots \mathfrak{q}_s,$$

then the right side lies in $\mathfrak{p}_1$. Primality forces some $\mathfrak{q}_j \subseteq \mathfrak{p}_1$, and maximality of nonzero primes gives equality. Cancel this invertible common factor and repeat.

Solution R10

The norm $N(a + b\sqrt{-5}) = a^2 + 5b^2$ shows the units are ± 1 and that $2, 3, 1 \pm \sqrt{-5}$ are irreducible because norms 2 and 3 do not occur. Thus

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$$

is genuinely nonunique at the element level.

Define

$$\mathfrak{p}_2 = (2, 1 + \sqrt{-5}), \quad \mathfrak{p}_3 = (3, 1 + \sqrt{-5}), \quad \bar{\mathfrak{p}}_3 = (3, 1 - \sqrt{-5}).$$

Direct computations give

$$(2) = \mathfrak{p}_2^2, \quad (3) = \mathfrak{p}_3 \bar{\mathfrak{p}}_3, \quad (1 + \sqrt{-5}) = \mathfrak{p}_2 \mathfrak{p}_3, \quad (1 - \sqrt{-5}) = \mathfrak{p}_2 \bar{\mathfrak{p}}_3.$$

Consequently both element factorizations induce

$$(6) = \mathfrak{p}_2^2 \mathfrak{p}_3 \bar{\mathfrak{p}}_3.$$

Unique factorization has not vanished; it has moved from elements to ideals.

Solution R11

If $\mathcal{O}_K$ is finite as a $\mathbb{Z}$ -module, then it is Noetherian because $\mathbb{Z}$ is Noetherian and submodules of finite modules are finitely generated. This alone does not make it Dedekind: one must also know that $\mathcal{O}_K$ is integrally closed in K and that every nonzero prime ideal is maximal (dimension one). Once those facts are established, the definition gives that $\mathcal{O}_K$ is Dedekind. The unique ideal factorization theorem then supplies a unique product of nonzero prime ideals for every nonzero ideal, in particular for each principal ideal $p\mathcal{O}_K$ generated by a rational prime.

Solution R12

A strong answer should connect the following ideas. Chebotarev concerns prime ideals in rings of integers, not merely rational primes. Ideals are needed because element factorization can fail; Dedekind domains restore unique factorization at the ideal level. For each nonzero prime $\mathfrak{p}$, localization produces the DVR $\mathcal{O}_{K,\mathfrak{p}}$, where divisibility is measured by a single valuation. In an extension L/K , a prime ideal of $\mathcal{O}_K$ factors into prime ideals of $\mathcal{O}_L$; the ramification indices are precisely local valuation data. Residue rings $\mathcal{O}_K/\mathfrak{p}$ are fields because nonzero primes are maximal. Galois automorphisms act on primes above $\mathfrak{p}$ and on these residue fields, which is what eventually permits the definition of decomposition groups, inertia groups, and Frobenius. Thus the commutative algebra in this phase supplies the objects, local structure, and factorization language in which Chebotarev can even be stated.

Marking guide

Score R1–R11 out of 8 points each and R12 out of 12, for a total of 100.

For an 8-point proof problem, use approximately:

- 2 points: correct definitions/setup;
- 2 points: identifies the decisive idea;
- 3 points: logically complete derivation;
- 1 point: clear explanation of why the result matters or what hypothesis is used.

For R12, award 2 points each for correctly connecting: ideals, Dedekind factorization, localization/DVRs, residue fields, extension-of-primes, and Galois/Frobenius motivation.

Mastery checkpoint

85–100 and no foundational gap: move to Phase 2. You may carry one small repair topic with you.

70–84: move on only if the lost points are computational or technical. Spend 2–4 days repairing any conceptual weakness in localization, integrality, or the Dedekind proof first.

Below 70, or inability to reconstruct R3/R4/R9: remain in Phase 1 for another week. Redo the relevant exercise clusters without notes, then retake only the failed readiness questions.

The score is secondary to the type of error. A forgotten algebraic manipulation is less serious than not knowing why localization or invertibility is being used.

Chapter 22

A five-week study schedule

This schedule assumes about 8–10 hours per week. Compress it if your Phase 0 diagnostic showed that the material is returning rapidly.

Week	Main topic	Required output before moving on
1	Ideals, quotients, prime/maximal ideals; Noetherianity	Prove the quotient criteria and Noetherian equivalences from memory. Solve I1–I12 and N1–N4.
2	Modules, Hilbert basis theorem; localization	Explain the universal property of localization without notes; compute $\mathbb{Z}_{(p)}$ and $k[x, y]_{(x)}$. Solve N5–N10 and L1–L12.
3	Integrality and integral closure	Reconstruct the determinant trick, transitivity of integrality, and $\text{UFD} \Rightarrow \text{integrally closed}$. Solve G1–G12.
4	DVRs and Dedekind domains	Prove the local DVR theorem and understand every hypothesis. Reconstruct why nonzero ideals are invertible. Solve D1–D9.
5	Unique ideal factorization; examples; synthesis	Work D10–D14 and S1–S8. Give a 20-minute oral explanation of the whole phase. Then take R1–R12 closed-book.

22.1 Suggested weekly rhythm

For a ten-hour week, use roughly:

- 2 hours reading and annotating definitions/theorems;
- 5 hours solving problems;
- 2 hours reconstructing proofs on blank paper;
- 1 hour updating your “Chebotarev map”: one page answering what each concept contributes to later number theory.

Chapter 23

Mastery checklist: when to move to Phase 2

Do not require perfection. Require functional command.

Definitions you should state accurately

You should be able to define without notes:

- ideal, prime ideal, maximal ideal;
- quotient ring;
- Noetherian ring;
- localization $S^{-1}A$;
- local ring;
- integral element;
- integral closure;
- PID and UFD;
- discrete valuation and DVR;
- Dedekind domain;
- fractional ideal;
- invertible ideal.

Proofs you should reconstruct

You need not reproduce the text verbatim, but you should independently rebuild:

1. prime/maximal quotient criteria;
2. Noetherian equivalences;
3. universal property and prime correspondence for localization;
4. finite-module criterion/determinant trick;
5. integrality is transitive;
6. UFDs are integrally closed;
7. principal maximal ideal + Noetherian local domain $\Rightarrow$ DVR;

8. Dedekind localization $A_{\mathfrak{p}}$ is a DVR;
9. every nonzero ideal of a Dedekind domain is invertible;
10. existence and uniqueness of prime ideal factorization.

Computations you should perform fluently

You should be able to:

- compute quotients and use the CRT;
- identify which elements become units after localization;
- compute prime ideals surviving a localization;
- test simple algebraic numbers for integrality;
- work with v_p in $\mathbb{Z}_{(p)}$;
- multiply, divide, intersect, and add factored ideals using prime exponents;
- explain the ideal factorization of 6 in $\mathbb{Z}[\sqrt{-5}]$.

Conceptual questions

You are ready to advance if you can answer, in your own words:

1. Why are ideals more robust than elements for factorization?
2. Why does localization at $\mathfrak{p}$ remove information about other primes?
3. Why is a monic polynomial the right condition in the definition of integrality?
4. Why should a DVR be regarded as arithmetic controlled by one prime?
5. Where does each of the three Dedekind hypotheses enter the theory?
6. Why does invertibility of ideals replace cancellation of nonzero integers?
7. Why will $p\mathcal{O}_K = \prod \mathfrak{p}_i^{e_i}$ be the natural starting point for studying splitting of primes?

Mastery checkpoint

Move on when you can answer at least six of the seven conceptual questions cleanly, reconstruct at least eight of the ten proof targets with at most minor gaps, and solve ordinary localization/integrality/ideal-factorization exercises without repeatedly consulting definitions. Do not delay Phase 2 because one technical Hilbert-basis-theorem detail is rusty; do delay if localization, integrality, or the Dedekind factorization proof still feels opaque.

Chapter 24

Compact reference sheet

Prime ideal. $\mathfrak{p}$ proper and $ab \in \mathfrak{p} \Rightarrow a \in \mathfrak{p}$ or $b \in \mathfrak{p}$; equivalently $A/\mathfrak{p}$ is a domain.

Maximal ideal. $\mathfrak{m}$ proper with no proper ideal strictly between $\mathfrak{m}$ and A ; equivalently $A/\mathfrak{m}$ is a field.

Noetherian. Every ideal finitely generated $\iff$ every ascending ideal chain stabilizes.

Localization. $S^{-1}A$ universally makes all $s \in S$ invertible.

At a prime. $A_{\mathfrak{p}} = (A \setminus \mathfrak{p})^{-1}A$ is local with maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$.

Integral element. x satisfies a monic polynomial over $A \iff A[x]$ is finite as an A -module.

Integrally closed. Elements of $\text{Frac}(A)$ integral over A already lie in A .

DVR. A local domain with a discrete valuation; every nonzero ideal is $\mathfrak{m}^n = (\pi^n)$.

Dedekind domain. Noetherian + integrally closed + every nonzero prime maximal.

Local structure. Dedekind A implies $A_{\mathfrak{p}}$ is a DVR for every nonzero prime $\mathfrak{p}$.

Fractional inverse. $I^{-1} = \{x \in K : xI \subseteq A\}$.

Dedekind ideals. Every nonzero fractional ideal is invertible and every nonzero ordinary ideal factors uniquely into nonzero prime ideals.

Exponent arithmetic. If $I = \prod \mathfrak{p}^{a_{\mathfrak{p}}}$, $J = \prod \mathfrak{p}^{b_{\mathfrak{p}}}$, then

$$IJ = \prod \mathfrak{p}^{a_{\mathfrak{p}}+b_{\mathfrak{p}}}, \quad I+J = \prod \mathfrak{p}^{\min(a_{\mathfrak{p}}, b_{\mathfrak{p}})}, \quad I \cap J = \prod \mathfrak{p}^{\max(a_{\mathfrak{p}}, b_{\mathfrak{p}})}.$$

What comes next

Phase 2 returns to field theory and Galois theory, now with the commutative-algebra language in place. The central targets will be finite extensions, separability and normality, the Galois correspondence, finite fields, and especially the Frobenius automorphism

$$x \longmapsto x^q$$

of finite fields. That Frobenius will later reappear as the residue-field automorphism attached to a prime ideal in a Galois extension.

The most important object to carry forward from this phase is not a theorem statement but the picture

$$\boxed{\text{global prime ideal } \mathfrak{p}} \longrightarrow \boxed{\text{local DVR } A_{\mathfrak{p}}} \longrightarrow \boxed{\text{one valuation } v_{\mathfrak{p}}}.$$

In algebraic number theory, this local picture will make ramification indices, residue fields, and eventually Frobenius elements natural rather than mysterious.

Recommended references for further depth

This workbook is self-contained for the Phase 1 objectives. If you want a second exposition, good companions are:

- J. S. Milne, *A Primer of Commutative Algebra*, especially localization, integral dependence, and Dedekind domains.
- Atiyah and Macdonald, *Introduction to Commutative Algebra*, especially Chapters 1, 3, 5, and 9 selectively.
- Dummit and Foote, *Abstract Algebra*, for a more elementary treatment of PIDs, UFDs, and modules.

Do not turn these into parallel cover-to-cover reading projects; use them only when a second explanation is useful.